

# Announcements

- Midterm Grades Released
  - 75% average for Section C
- HW3 ongoing

# Regularized Linear Regression

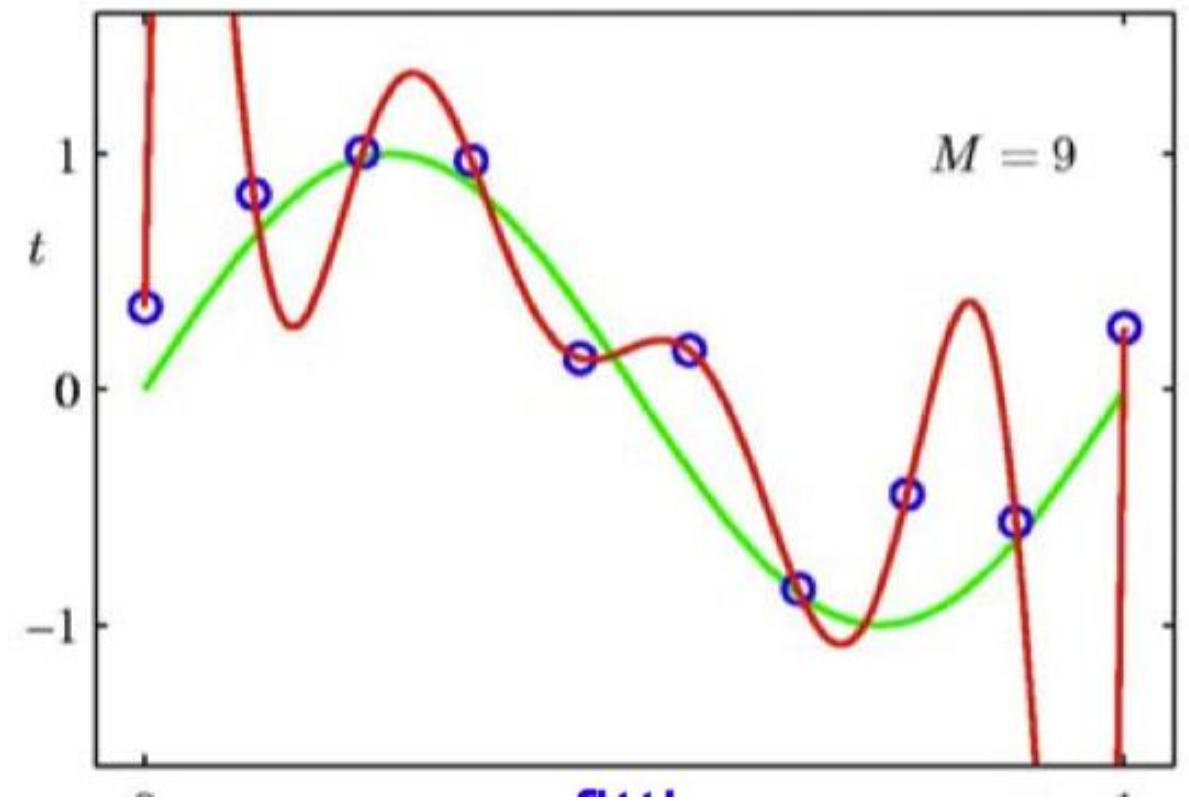
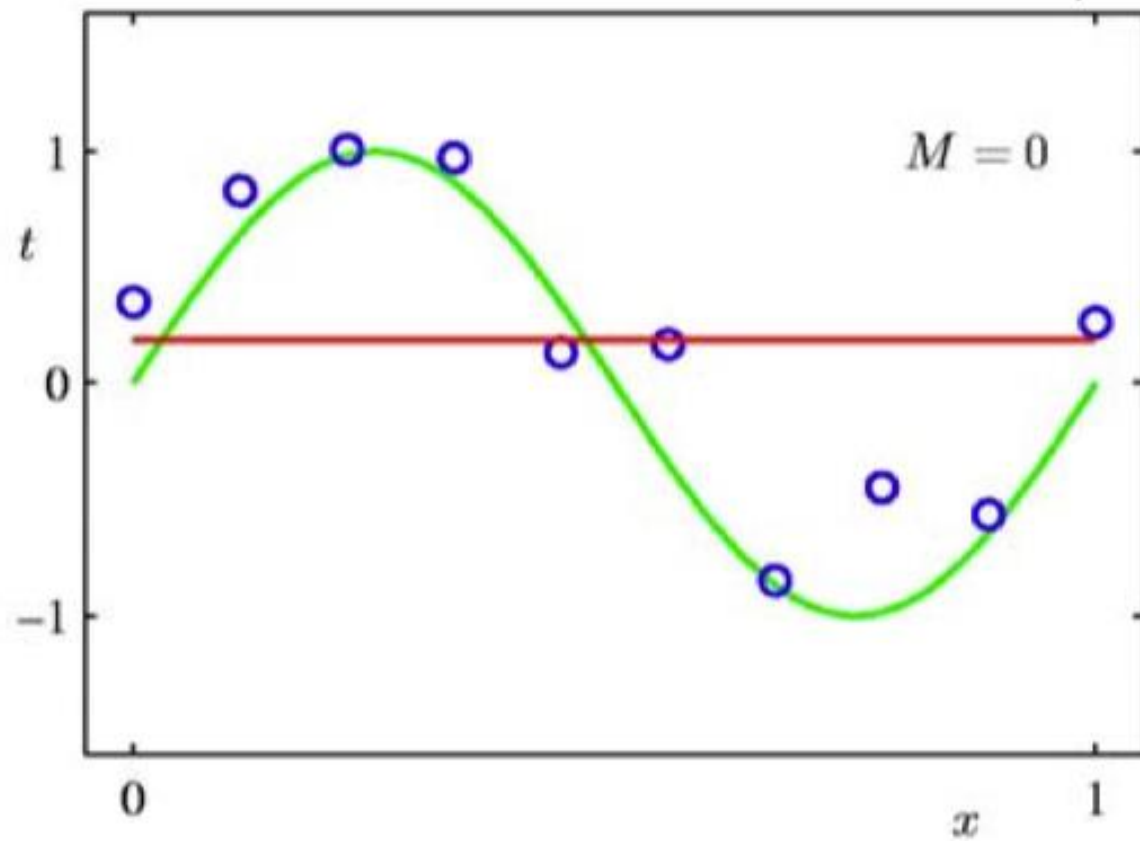
Nimisha Roy  
Georgia Tech

# Outline

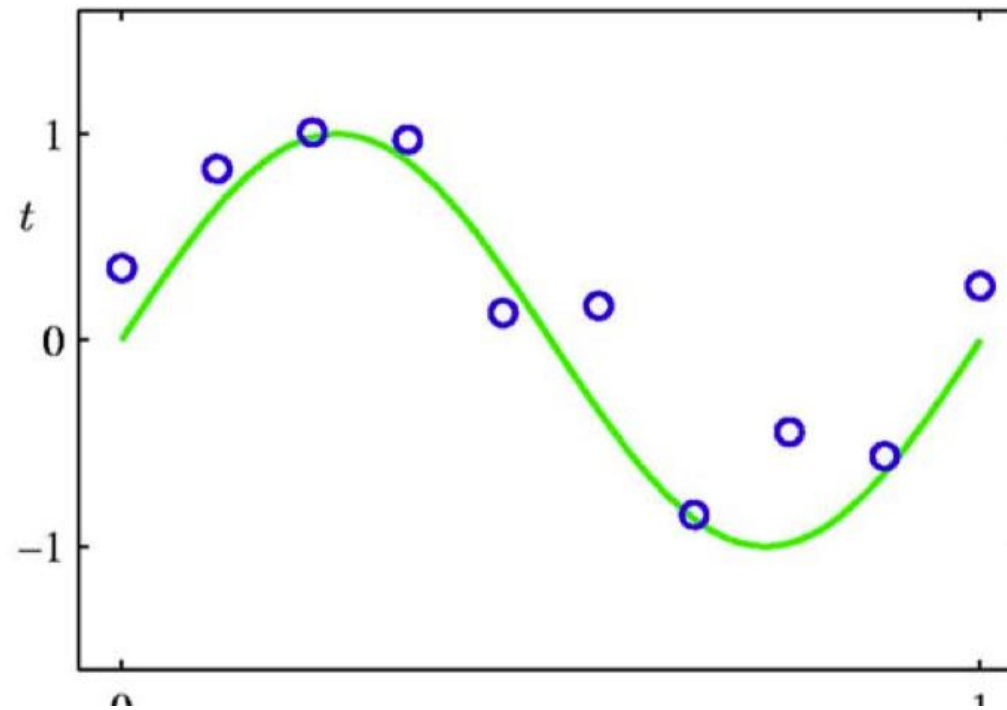
- Overfitting and regularized learning
- Ridge regression
- Lasso regression
- Determining regularization strength



# Recap: Overfitting vs Underfitting



# Regression: Why is overfitting more common?



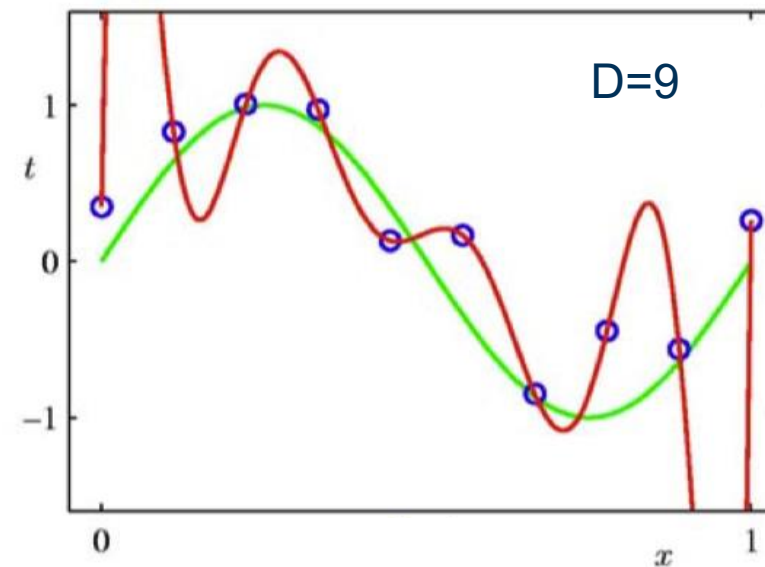
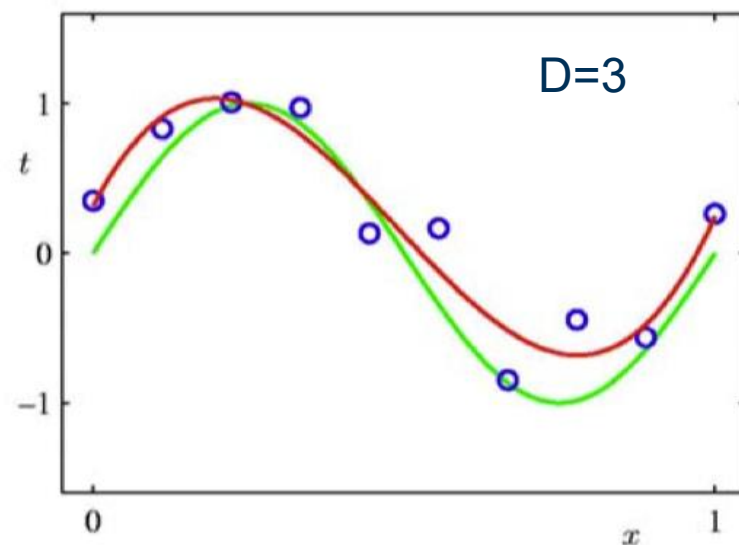
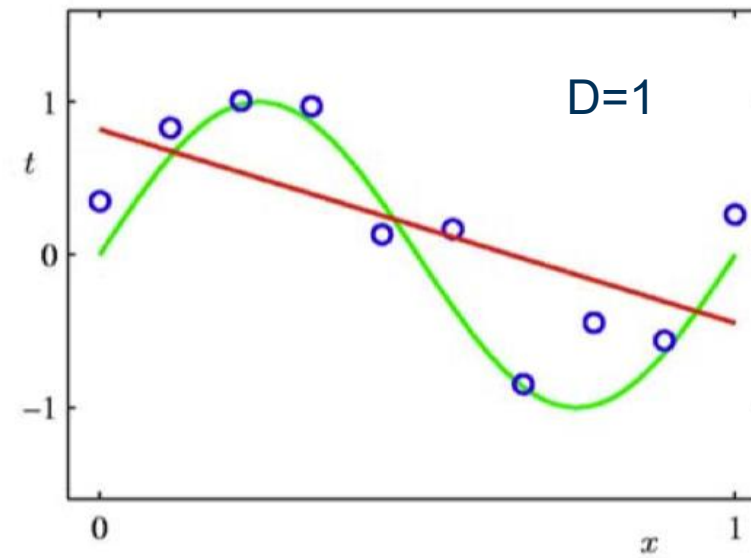
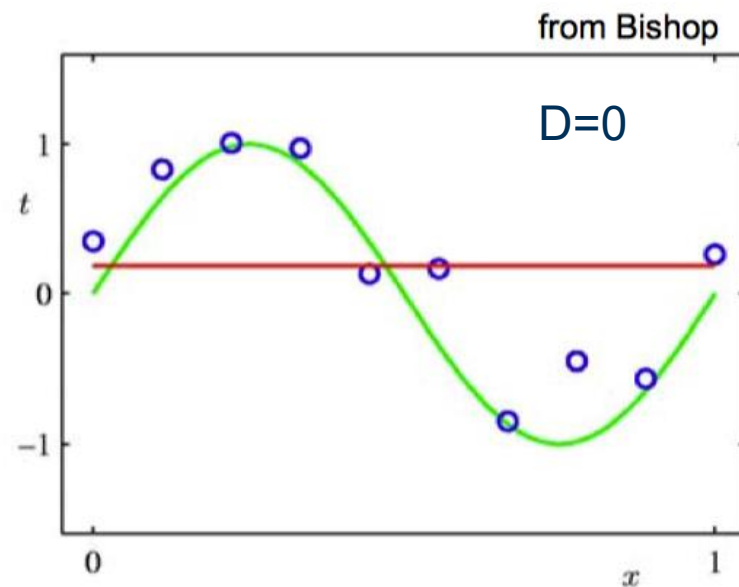
- Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_d x^d + \epsilon$$

- $z = \{1, x, x^2, \dots, x^d\} \in R^d$  and  $\theta = (\theta_0, \theta_1, \theta_2, \dots, \theta_d)^T$

$$y = z\theta$$

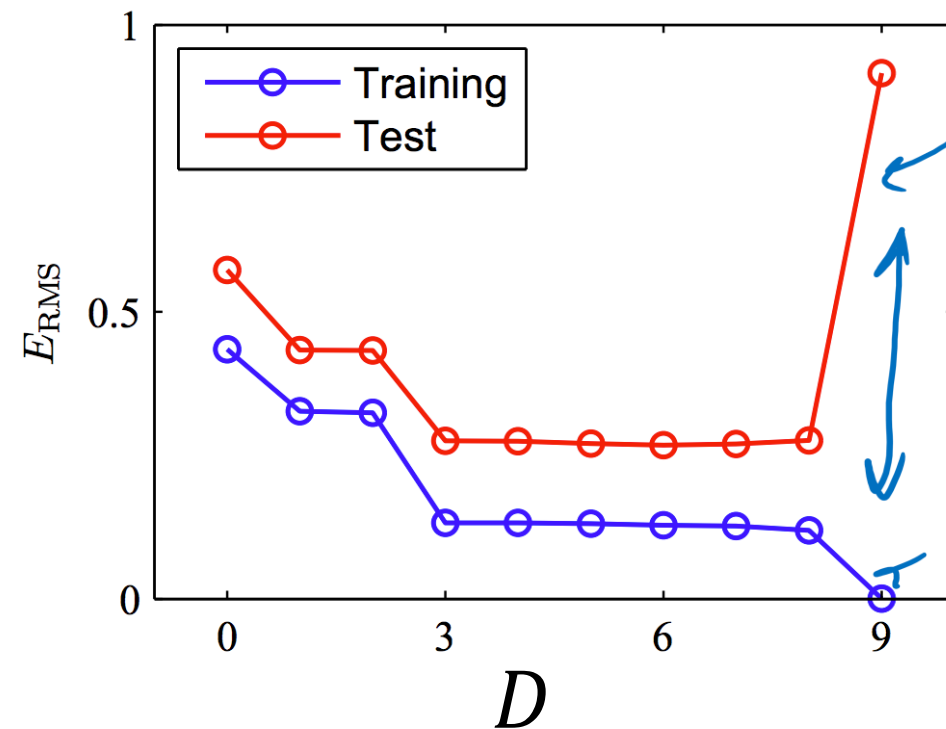
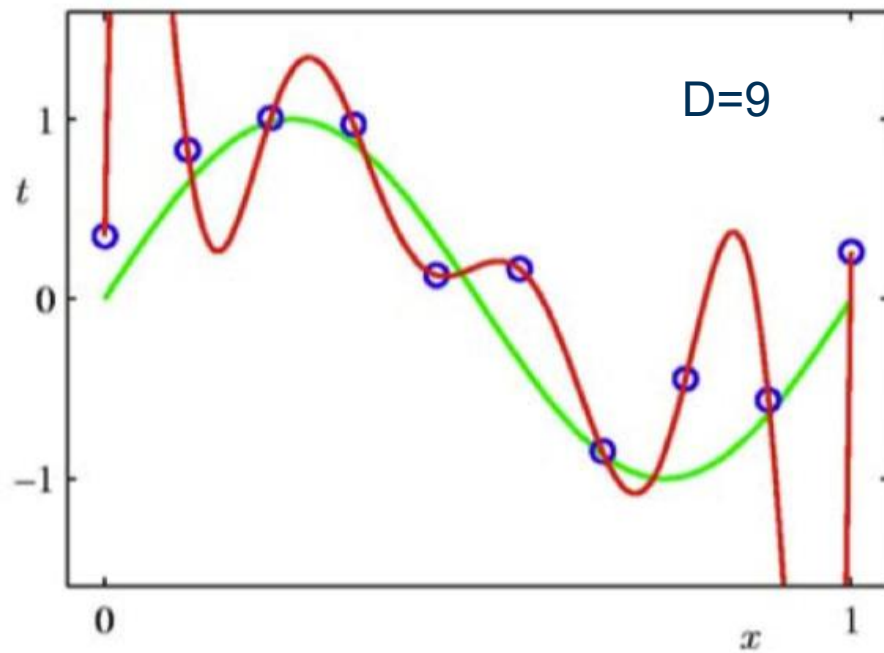
# Which One is Better?



- Can we increase the maximal polynomial degree to very large, such that the curve passes through all training points?

No, this can lead to **overfitting**!

# The Overfitting Problem

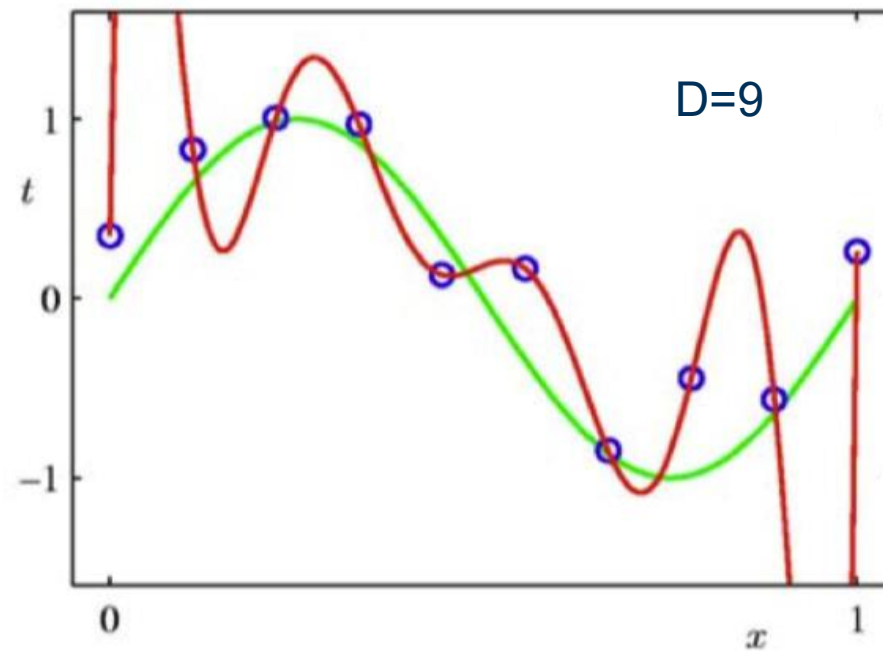


- The training error is very low, but the error on test set is large.
- The model captures not only patterns but also noisy nuisances in the training data.

# Root cause of overfitting?

- I want to convert 1 feature in  $x$  space to  $d$  features in  $z$  space

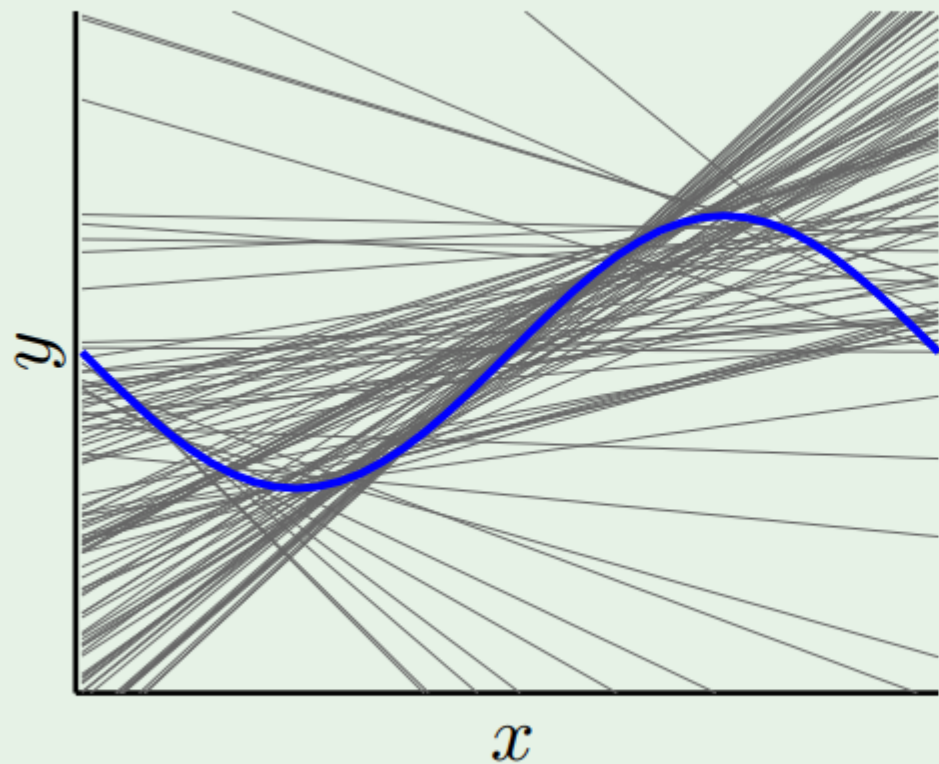
# The Overfitting Problem



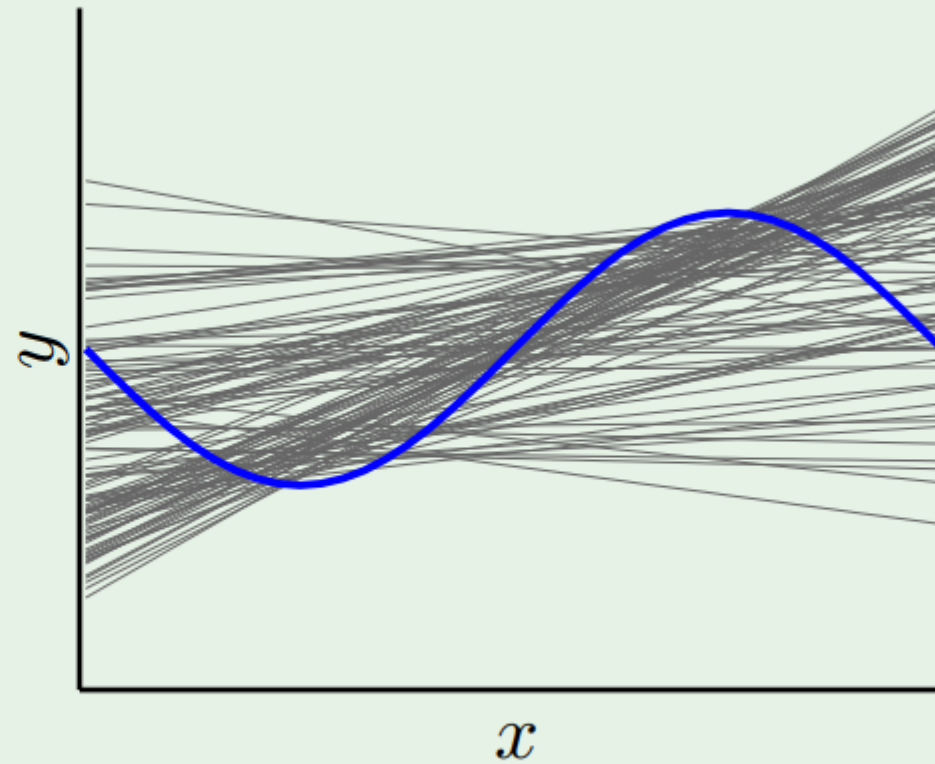
- In regression, overfitting is often associated with large Weights (severe oscillation)
- How can we address overfitting?

# Regularization

(smart way to cure overfitting disease )



without regularization



with regularization

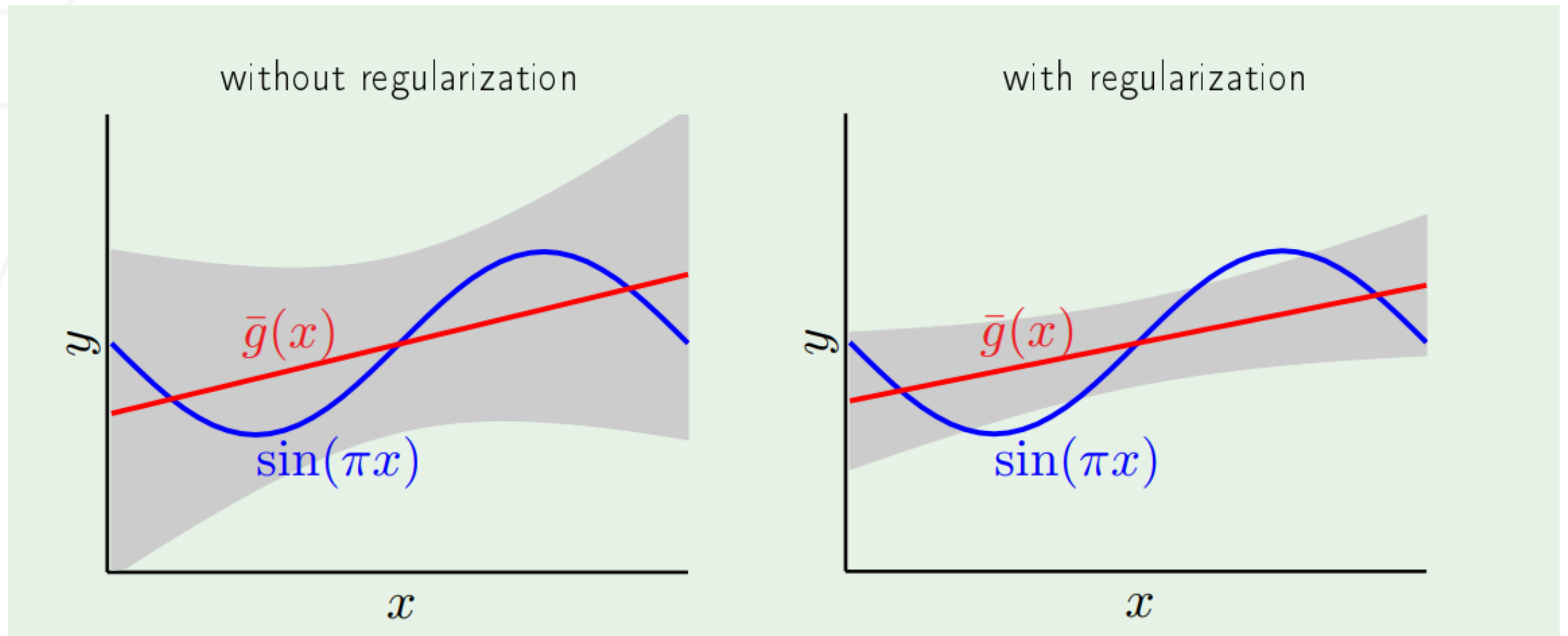
Put a brake on fitting



Fit a linear line on sinusoidal with just two data points

# Who is the winner?

$\bar{g}(x)$ : average over all lines



bias=0.21; var=1.69

bias=0.23; var=0.33

# So, what is regularization in general?

Regularization adds a penalty term to the error or objective function:

- The penalty discourages large weights → the model becomes *simpler*.
- A simpler model can't bend perfectly to fit all training data → **bias increases slightly**.
- But since it's not so sensitive to the quirks of any one dataset → **variance decreases significantly**.
- That tradeoff leads to better **generalization**.

| Regularization                                                                             | Bias            | Variance          | Total Error | Stability          |
|--------------------------------------------------------------------------------------------|-----------------|-------------------|-------------|--------------------|
|  Without | Low             | <b>High</b>       | High        | Overfits           |
|  With   | Slightly higher | <b>Much lower</b> | Lower       | Generalizes better |

# Polynomial Model

Want to fit a polynomial regression model

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \cdots + \theta_d x^d + \epsilon$$

Let's rewrite it as:

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d + \epsilon = \mathbf{z}\boldsymbol{\theta}$$

# Regularizing is just constraining the weights ( $\theta$ )

For example: let's do a **hard** constraining

$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d$$

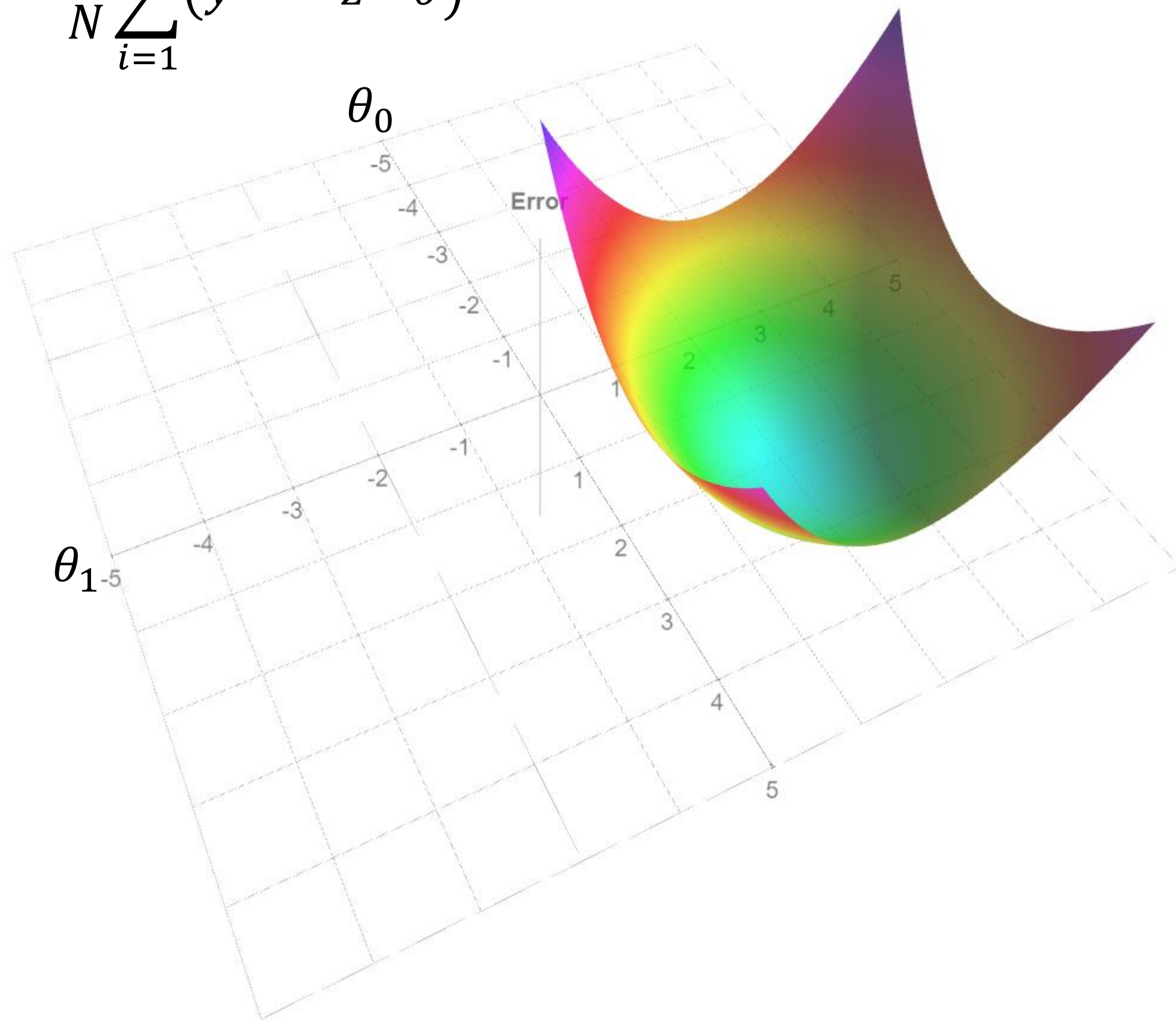
subject to

$$\theta_d = 0 \text{ for } d > 2$$

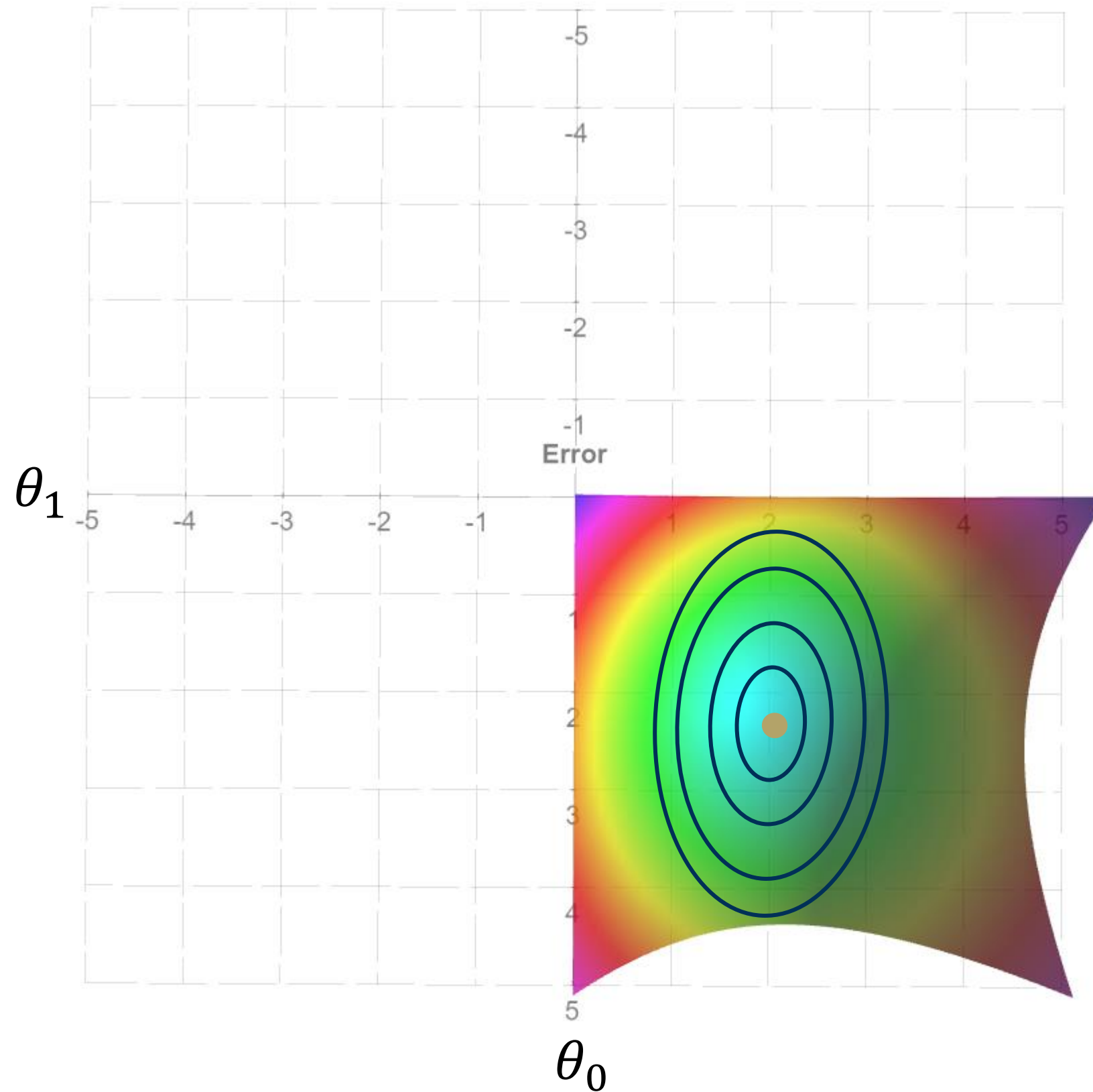


$$y = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + 0 + \cdots + 0$$

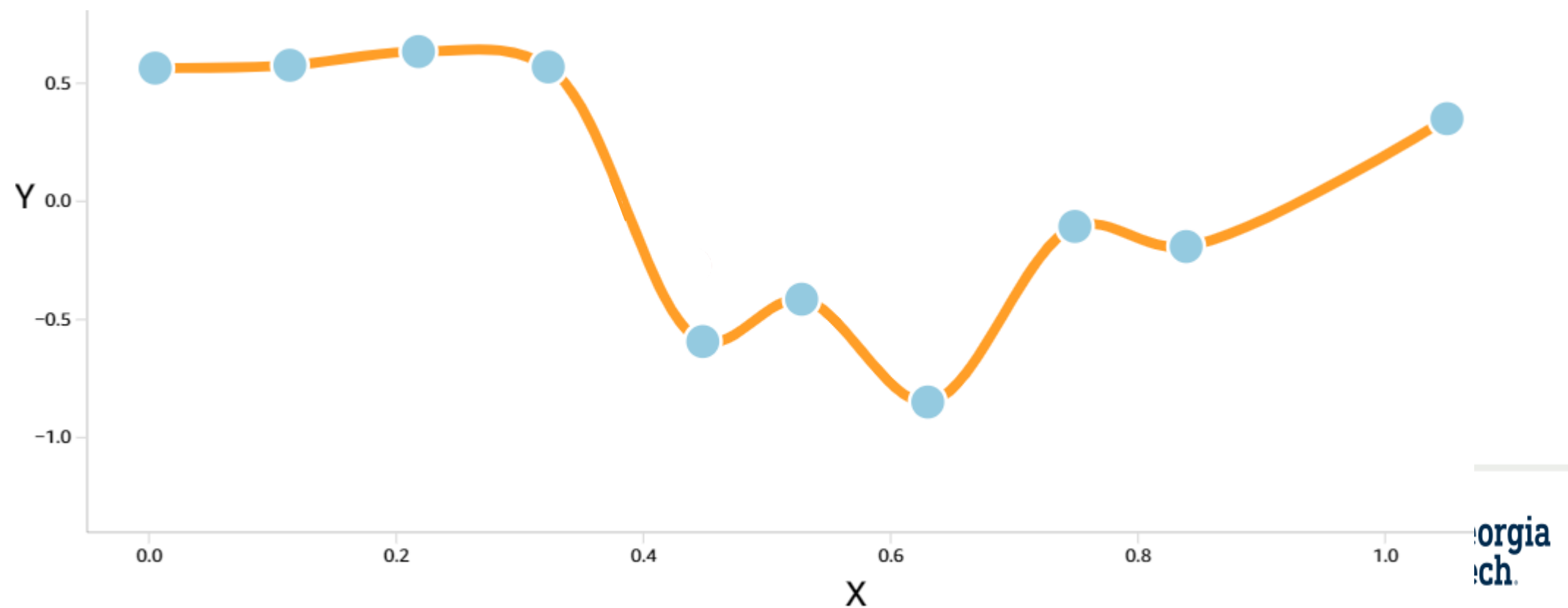
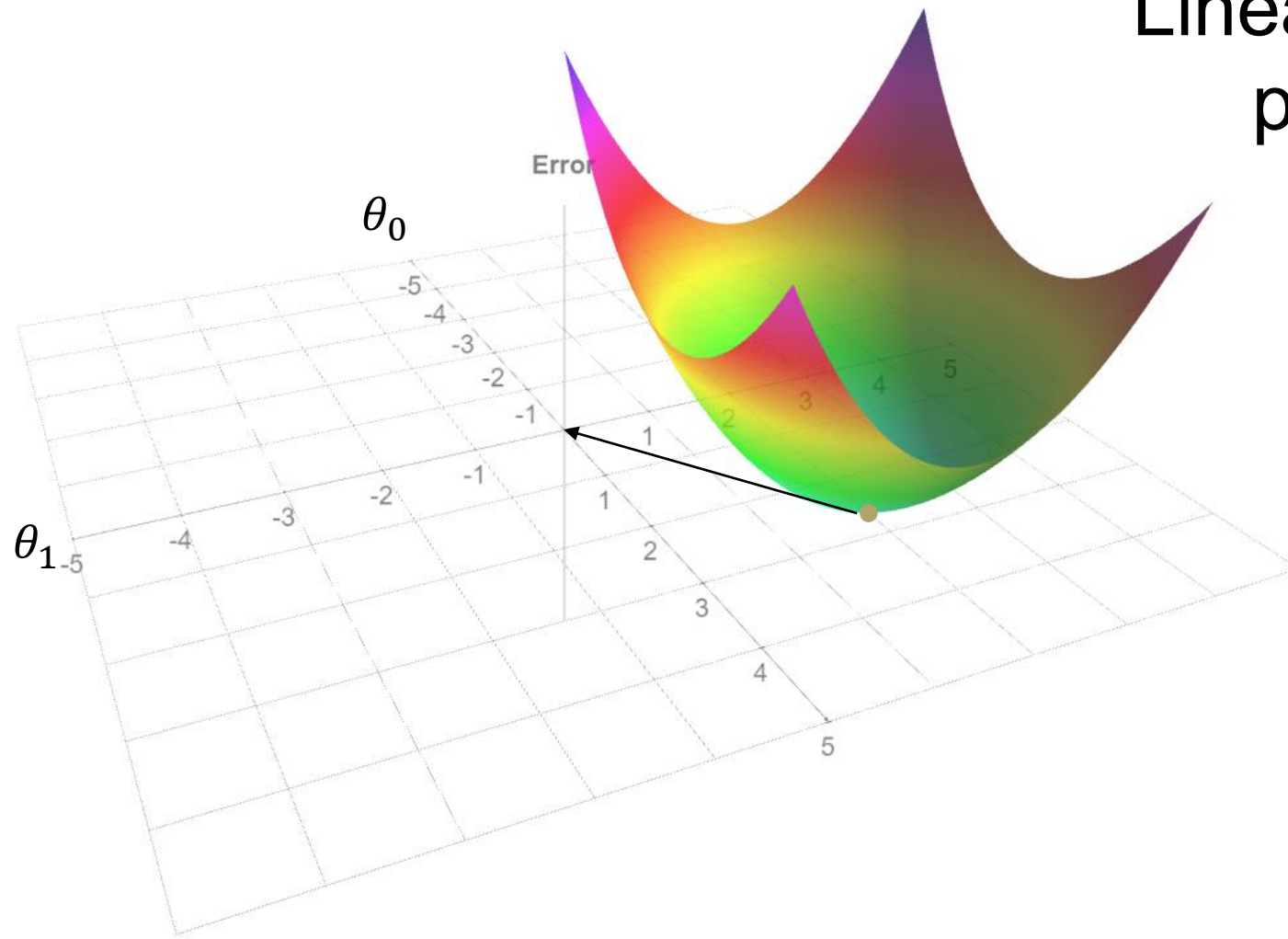
$$E(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$

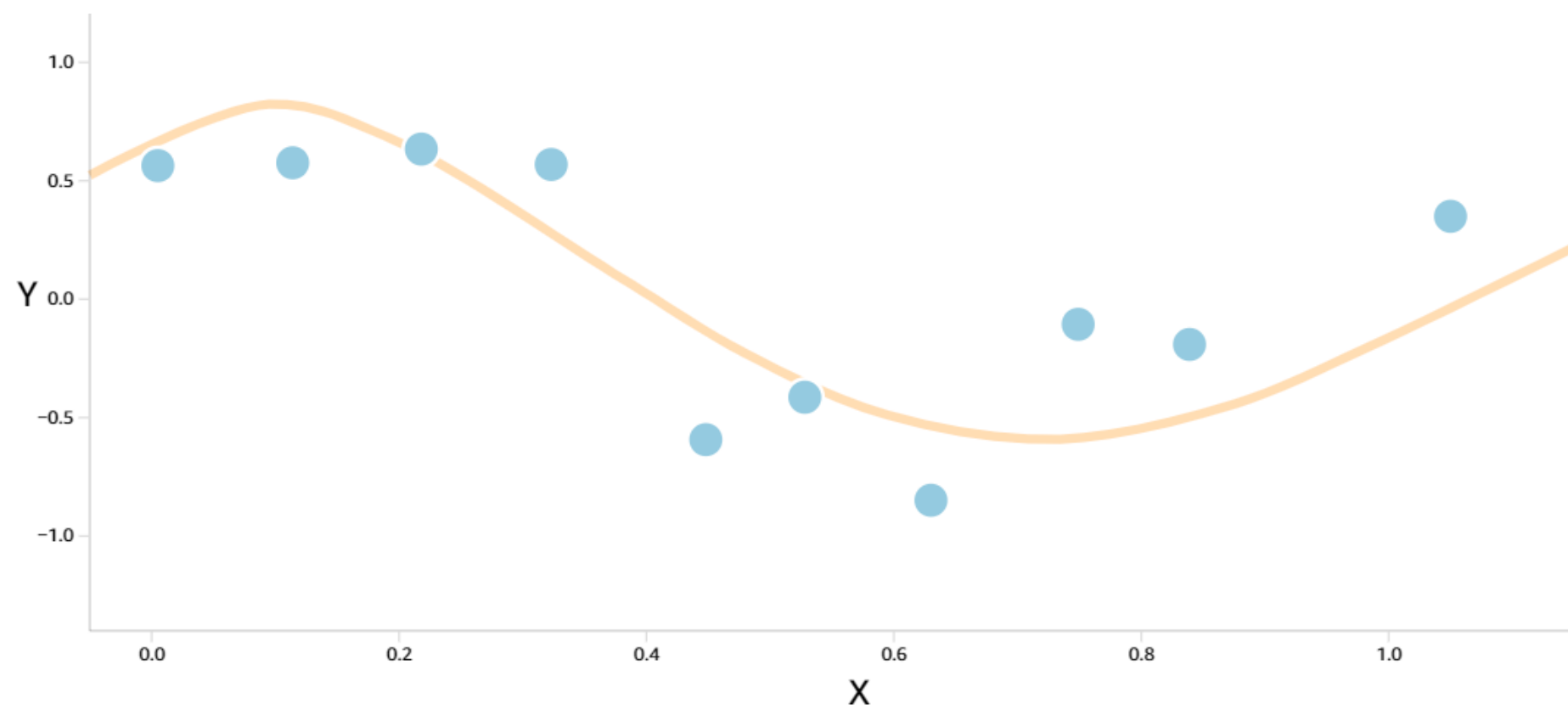
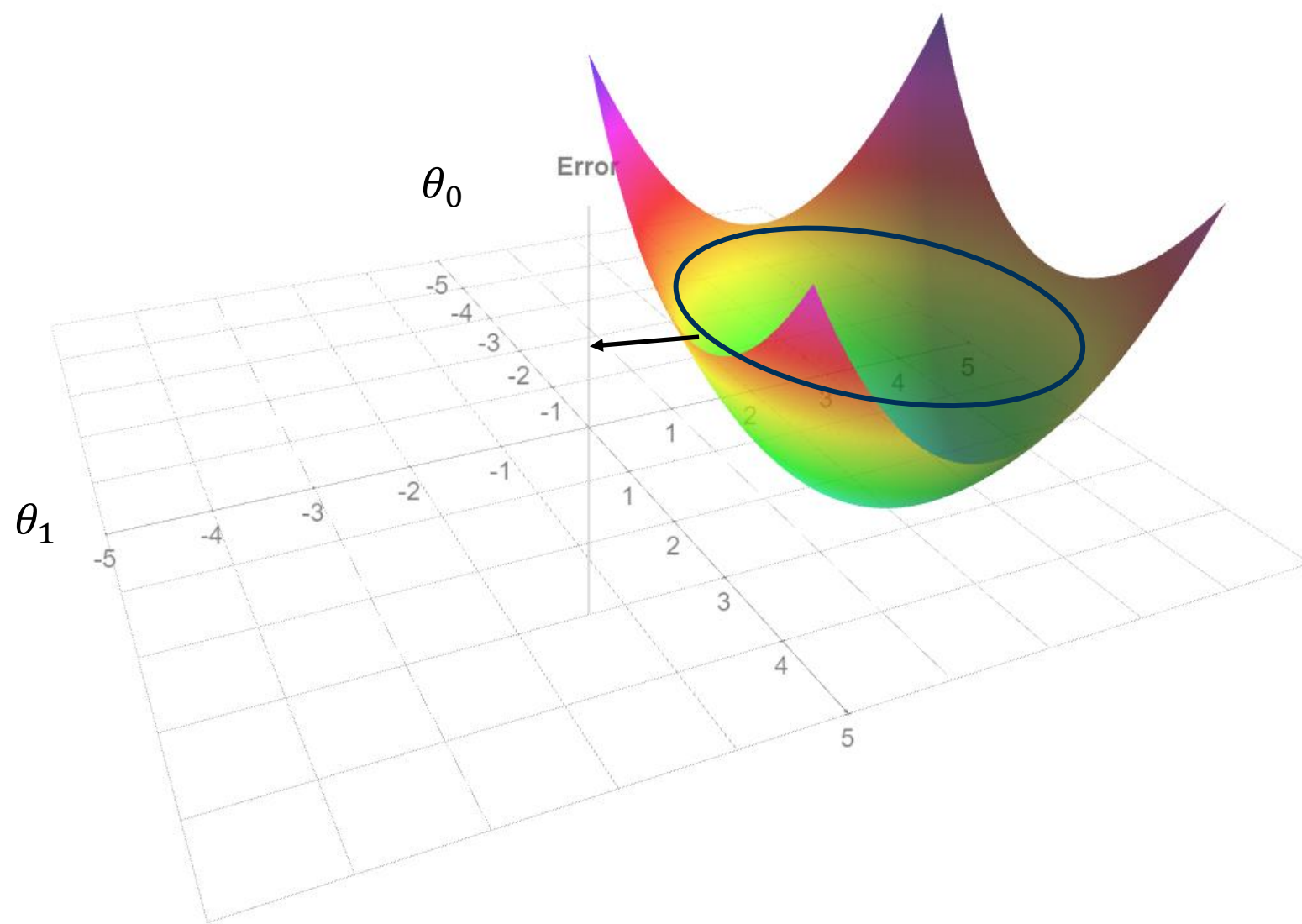


# Project the same graph on x-y using contour plot



# Linear regression with a very high polynomial degree solution



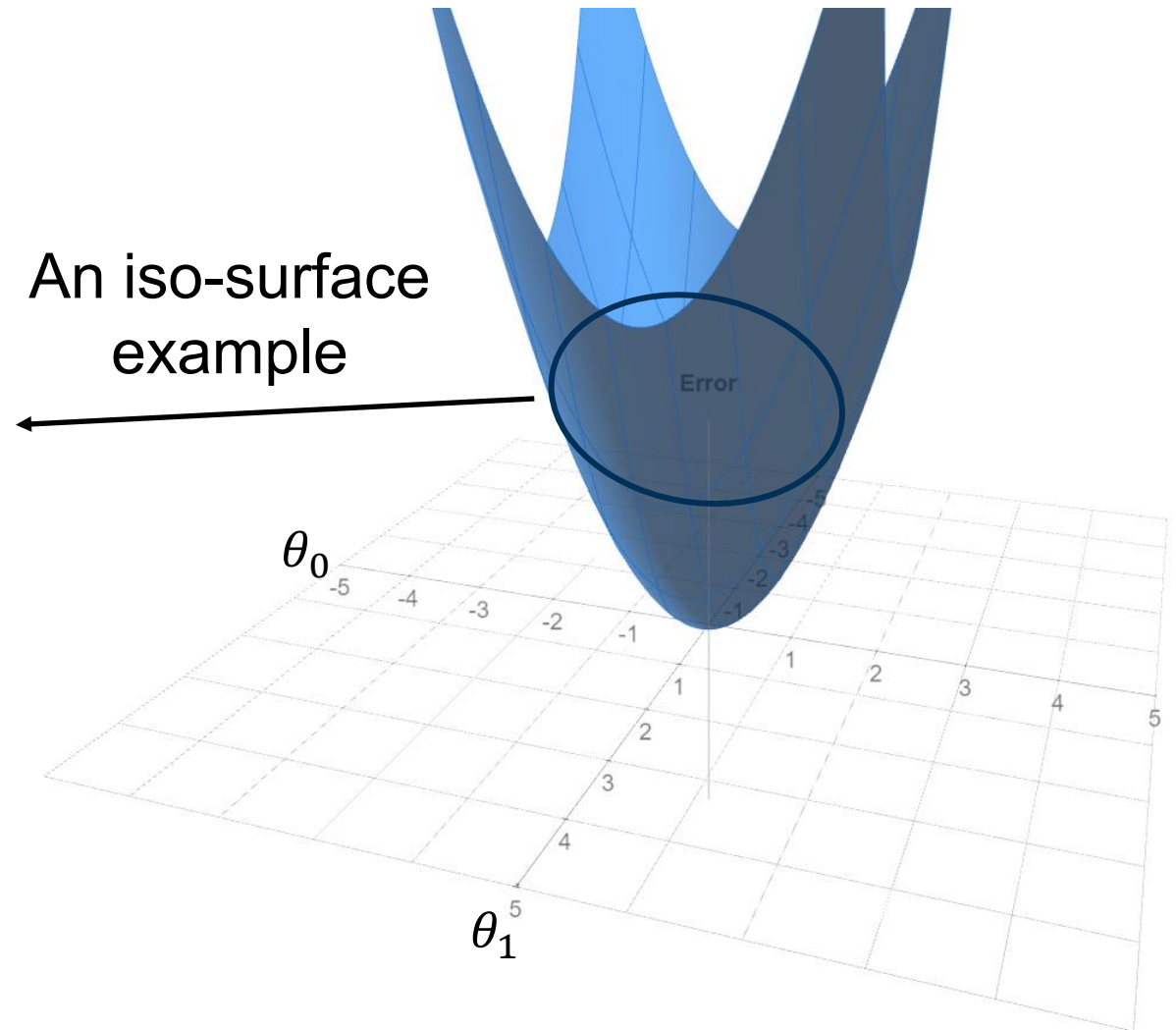


# How can we get an optimal solution with a positive error for a model that overfits?

We need to introduce a constraint

$$g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta = C$$

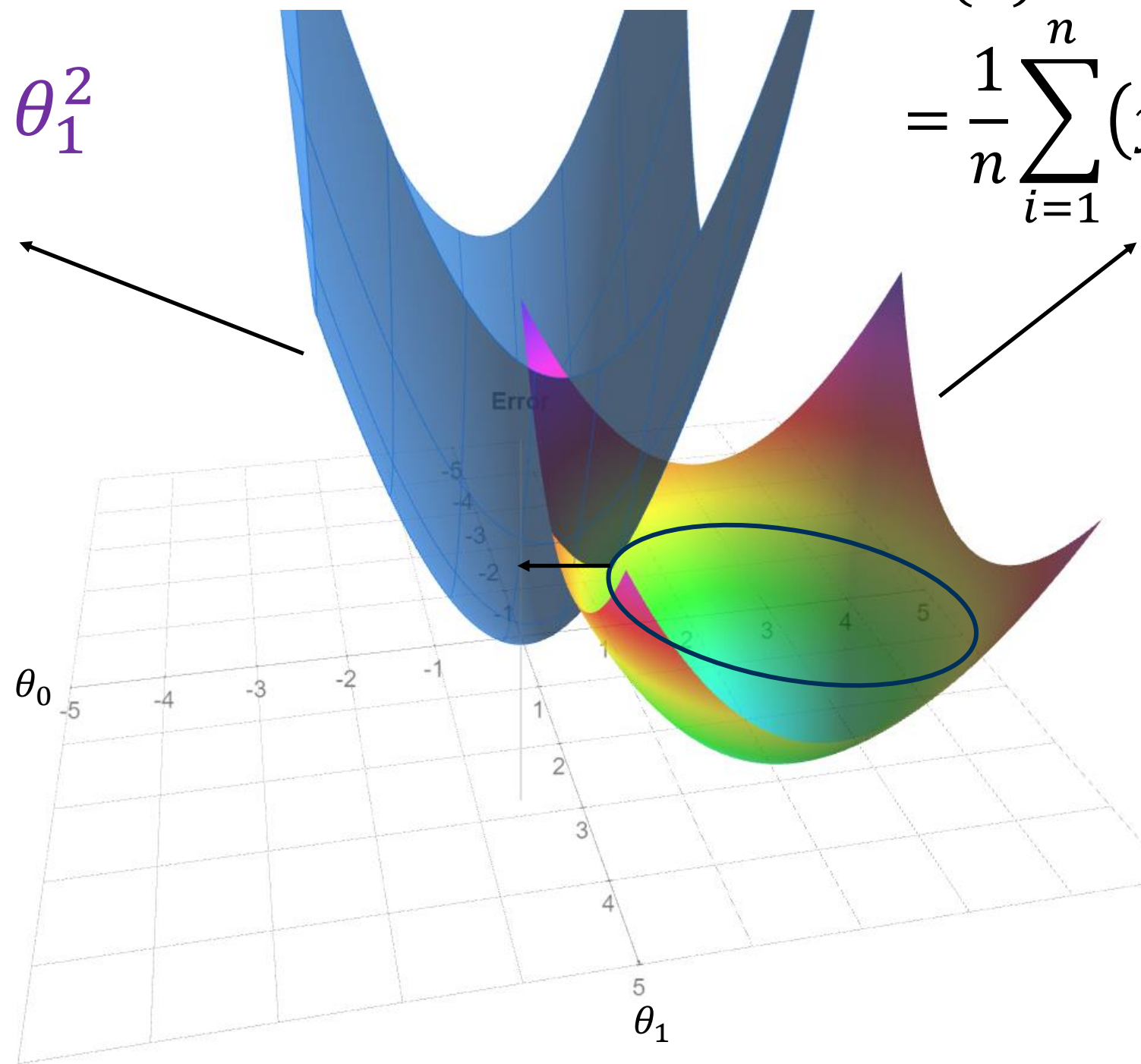
An iso-surface example



# Error function together with a new introduced constraint

$$g(\theta) = \theta_0^2 + \theta_1^2 \\ = \theta^T \theta$$

$$E(\theta) \\ = \frac{1}{n} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$



## Let's define the Lagrange function

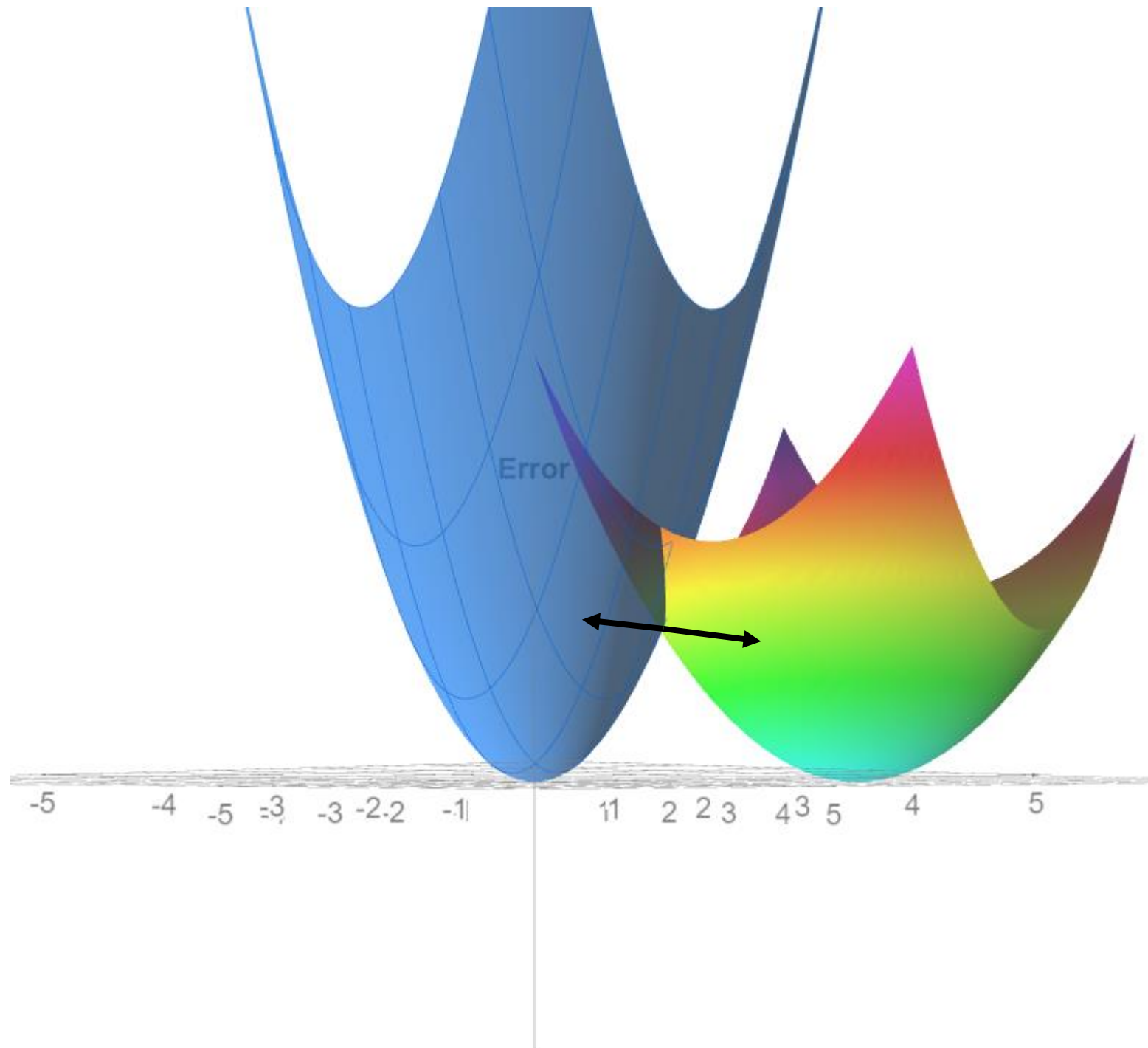
$$L(\theta, \lambda) = E(\theta) + \lambda g(\theta)$$

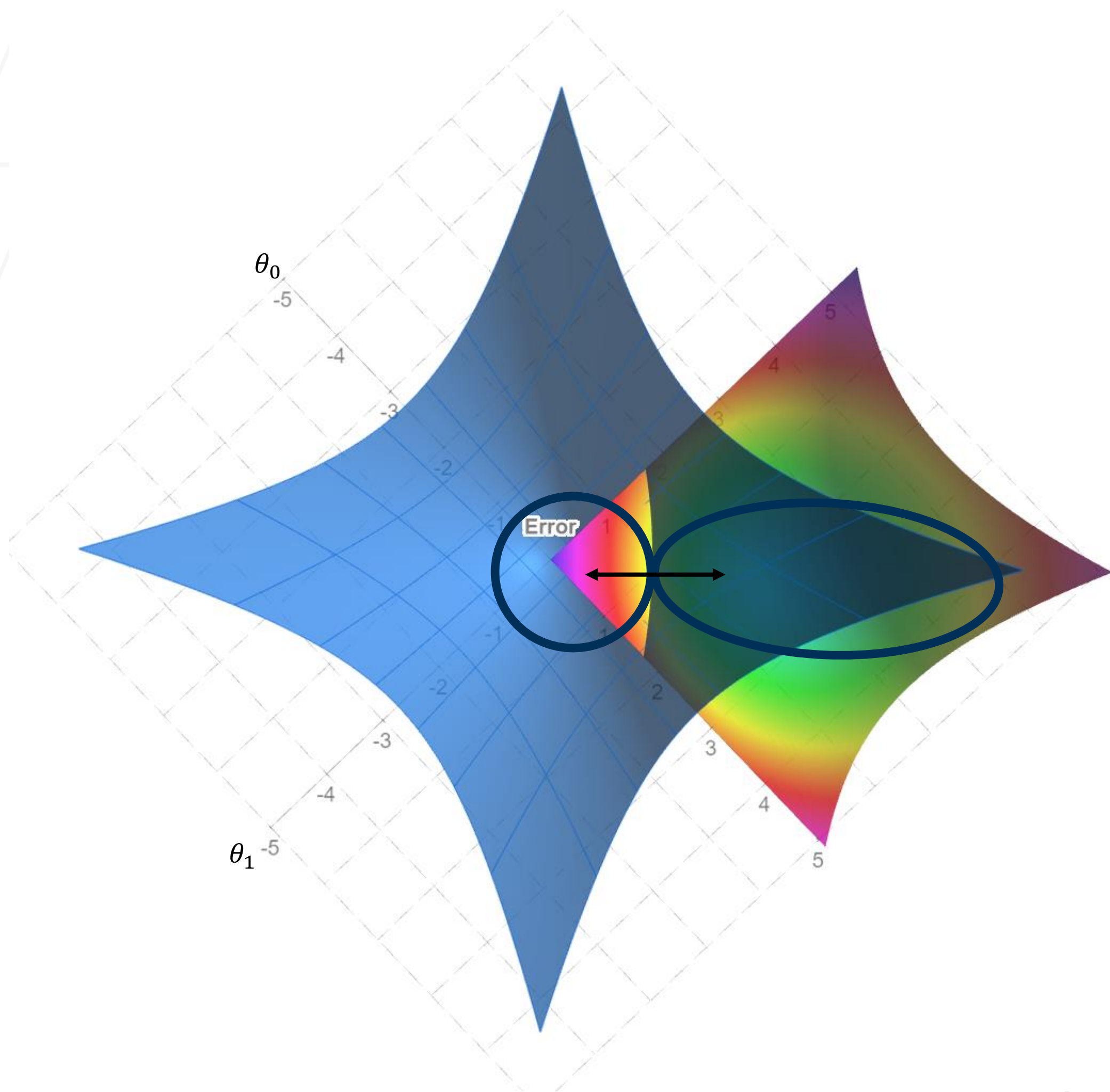
$$L(\theta, \lambda) = E(\theta) + \lambda \theta^T \theta - \lambda C$$

$$\nabla L(\theta, \lambda) = 0 \quad \nabla [E(\theta) + \lambda \theta^T \theta - \lambda C] = 0$$

$$\nabla [E(\theta)] + \lambda \nabla [\theta^T \theta] = 0$$

# How to enforce the gradient of Lagrange function to be zero







# Let's calculate the gradients

Gradient of constraint  $g(\theta)$       $\nabla[\theta^T \theta] = 2\theta$

$$\nabla[E(\theta)] + \lambda \nabla[\theta^T \theta] = 0$$

$$\nabla[E(\theta)] = -\lambda \nabla[\theta^T \theta]$$

$$\nabla E(\theta) = -2\lambda\theta$$

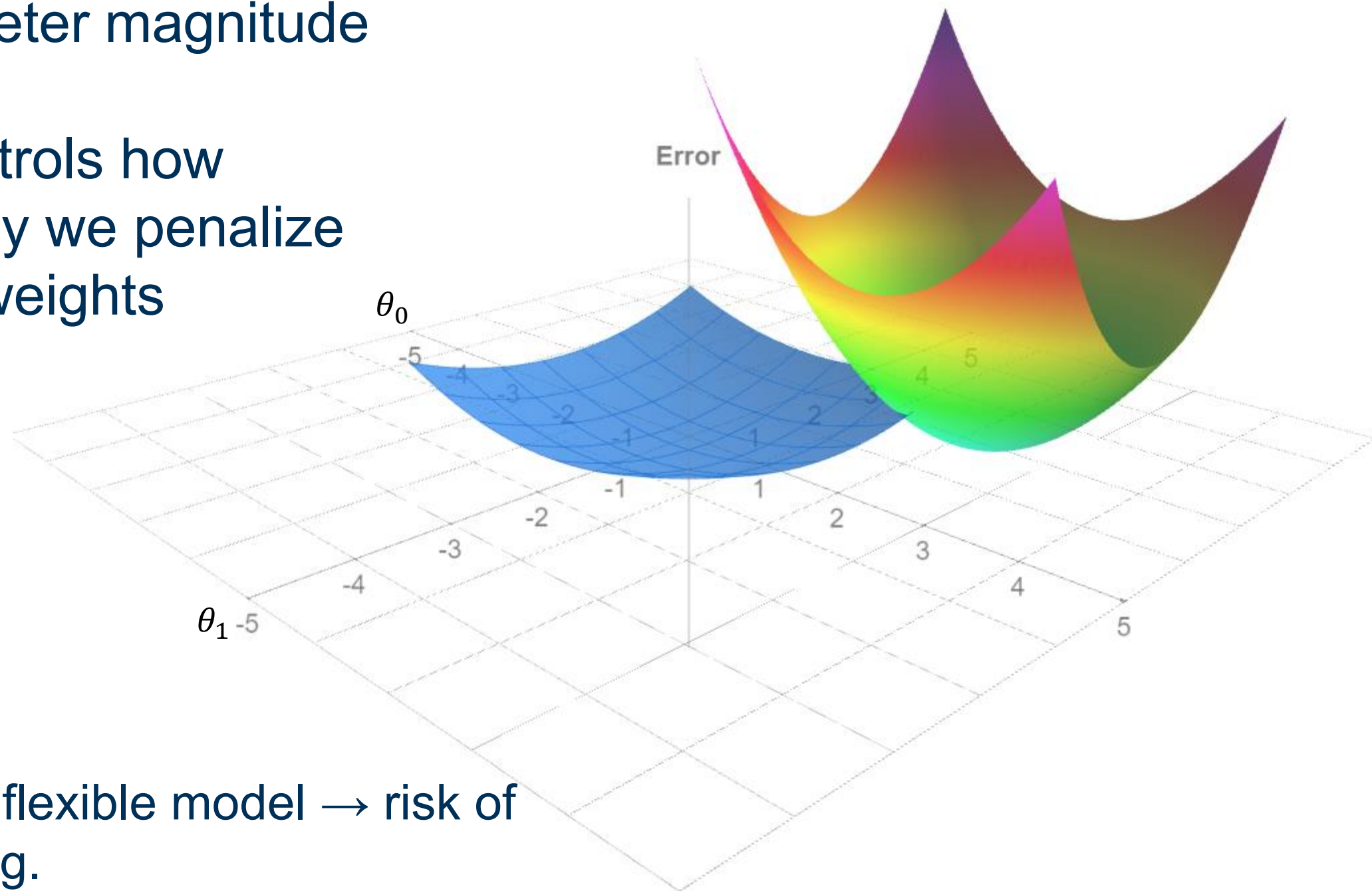
$$\left. \nabla E(\theta) + 2\lambda\theta = 0 \right\} \text{ Let's do integration } E(\theta) + \lambda\theta^T \theta$$

# The effect of low Lambda

$\frac{\lambda}{N} \theta^T \theta$  : The **regularization penalty** on parameter magnitude

$\lambda$ : Controls how strongly we penalize large weights

$$E(\theta) + \frac{\lambda}{N} \theta^T \theta$$



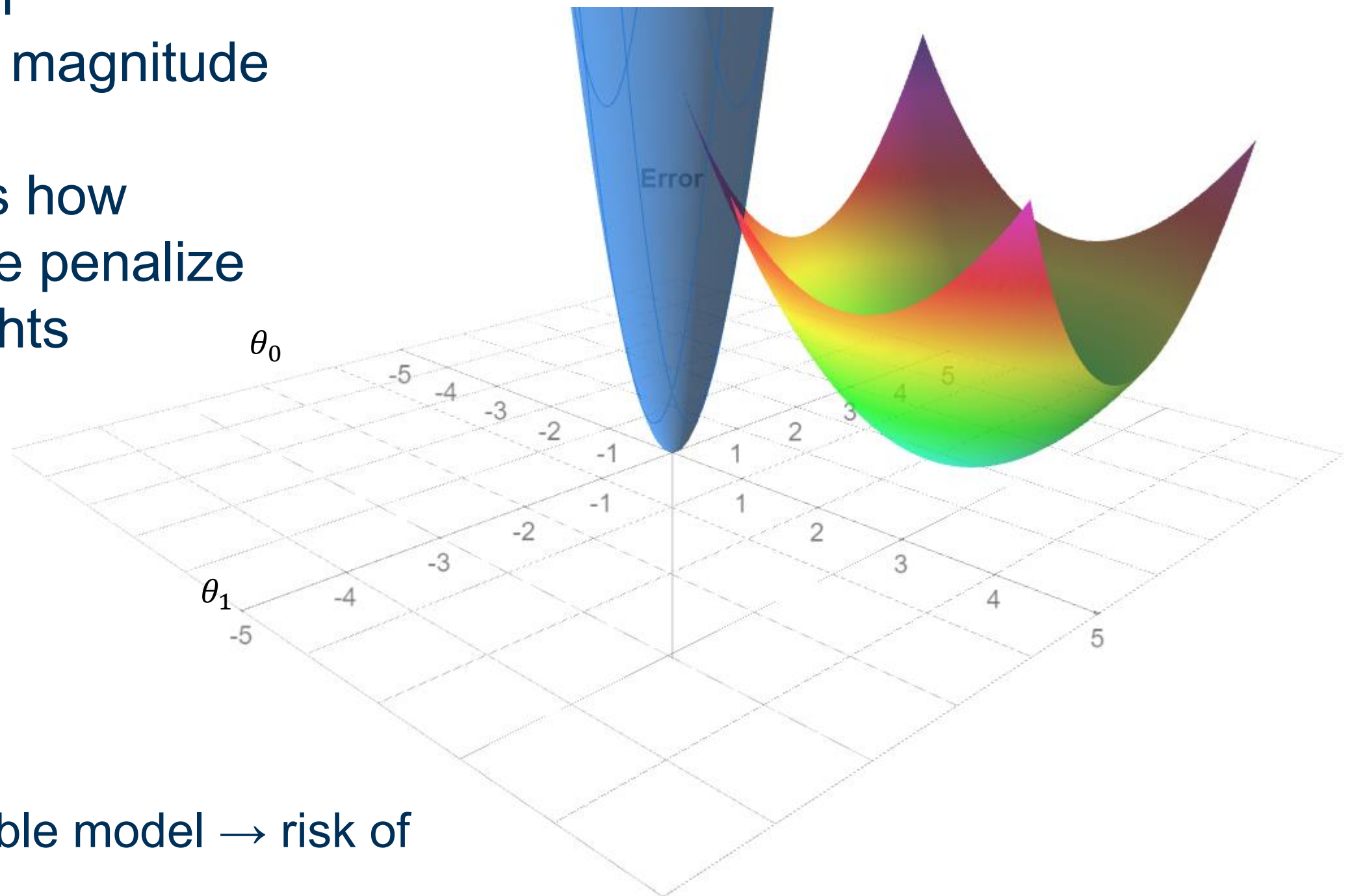
low  $\lambda \rightarrow$  flexible model  $\rightarrow$  risk of overfitting.

# The effect of high Lambda

$\frac{\lambda}{N} \theta^T \theta$  : The **regularization penalty** on parameter magnitude

$\lambda$ : Controls how strongly we penalize large weights

$$E(\theta) + \frac{\lambda}{N} \theta^T \theta$$



high  $\lambda \rightarrow$  stable model  $\rightarrow$  risk of underfitting.

# Regularized Learning

Minimize  $E(\theta) + \lambda \theta^T \theta$

Now we know Why this term leads to the regularization of parameters

Regularized Error

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \frac{\lambda}{2N} \|\theta\|_2^2$$

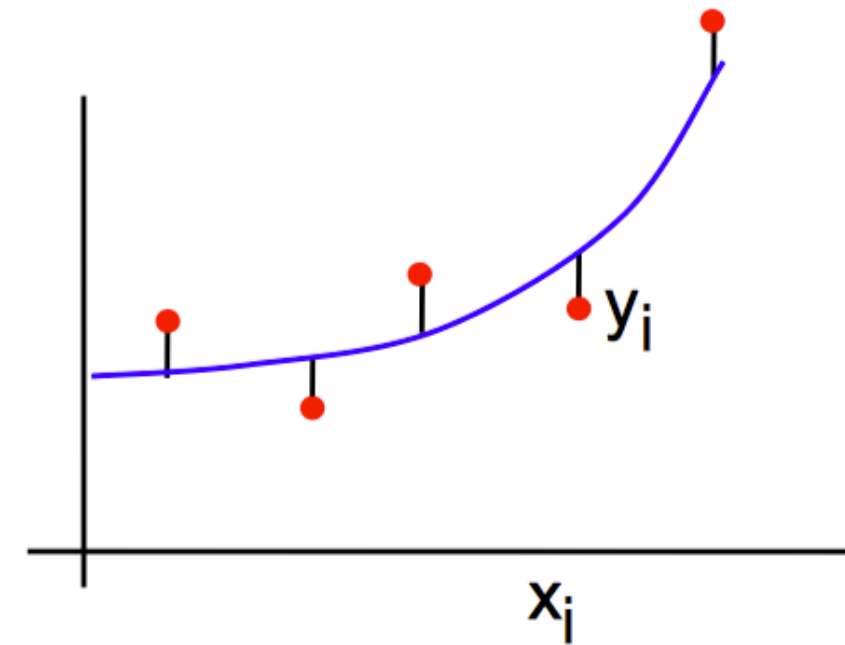
L2 Regularization term

# Outline

- Overfitting and regularized learning
- Ridge regression 
- Lasso regression
- Determining regularization strength

# Ridge Regression

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \frac{\lambda}{2N} \|\theta\|_2^2$$



$$\theta_0 + \theta_1 z_1 + \theta_2 z_2 + \cdots + \theta_d z_d + \epsilon = \mathbf{z} \theta$$

General form

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2$$

Matrix form

$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_2^2$$

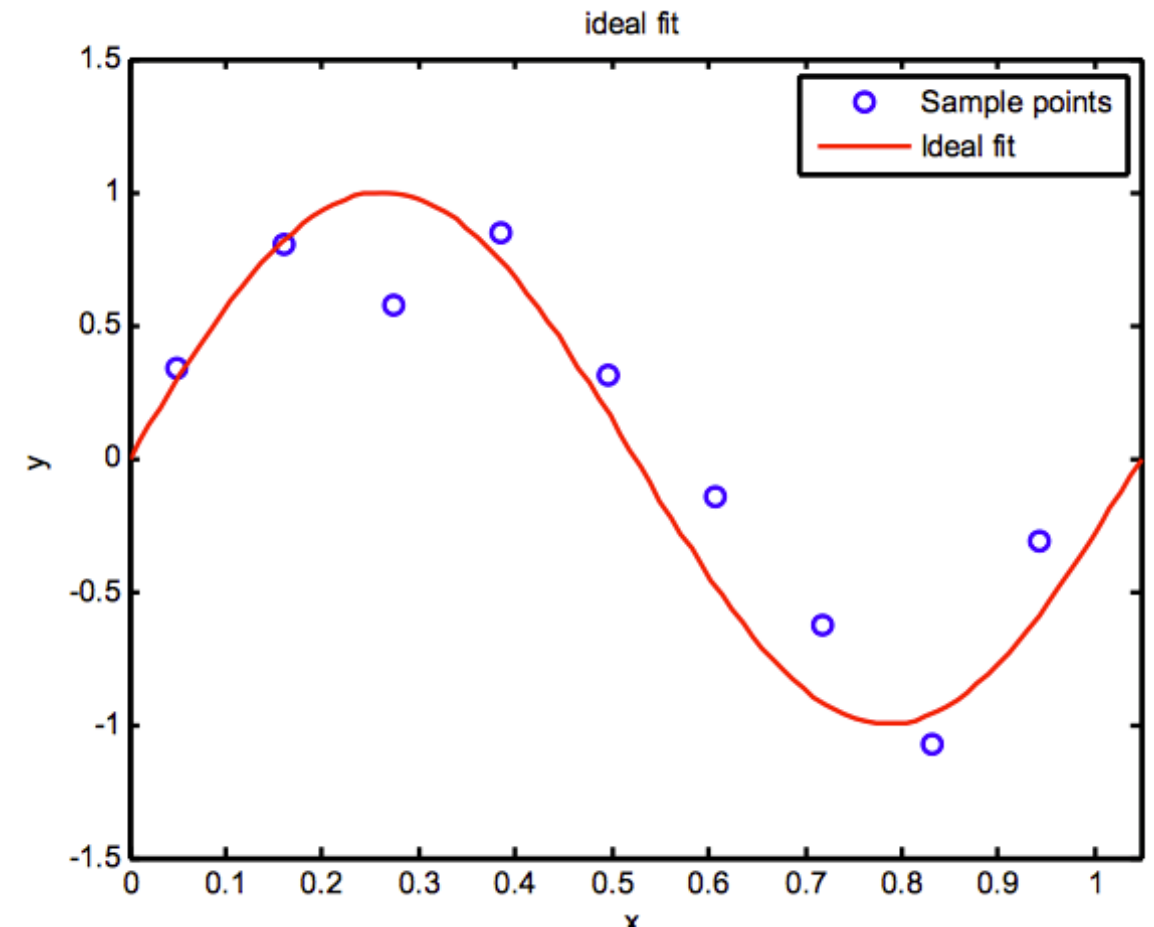
$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \lambda \theta$$

$$(z^T z + \lambda I) \theta = z^T y$$

$$\theta = (z^T z + \lambda I)^{-1} z^T y$$

# Ridge Regression Example

- The red curve is the true function (which is not a polynomial)
- The data points are samples from the curve with added noise in  $y$ .
- There is a choice in both the degree,  $D$ , of the basis functions used, and in the strength of the regularization

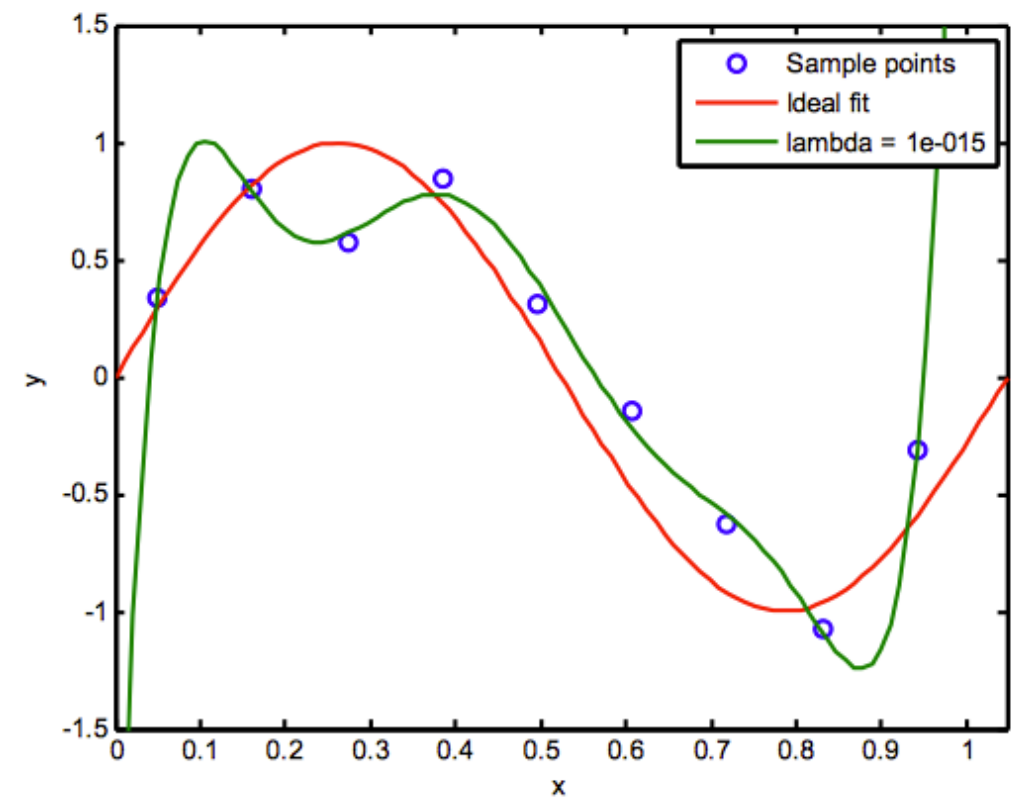
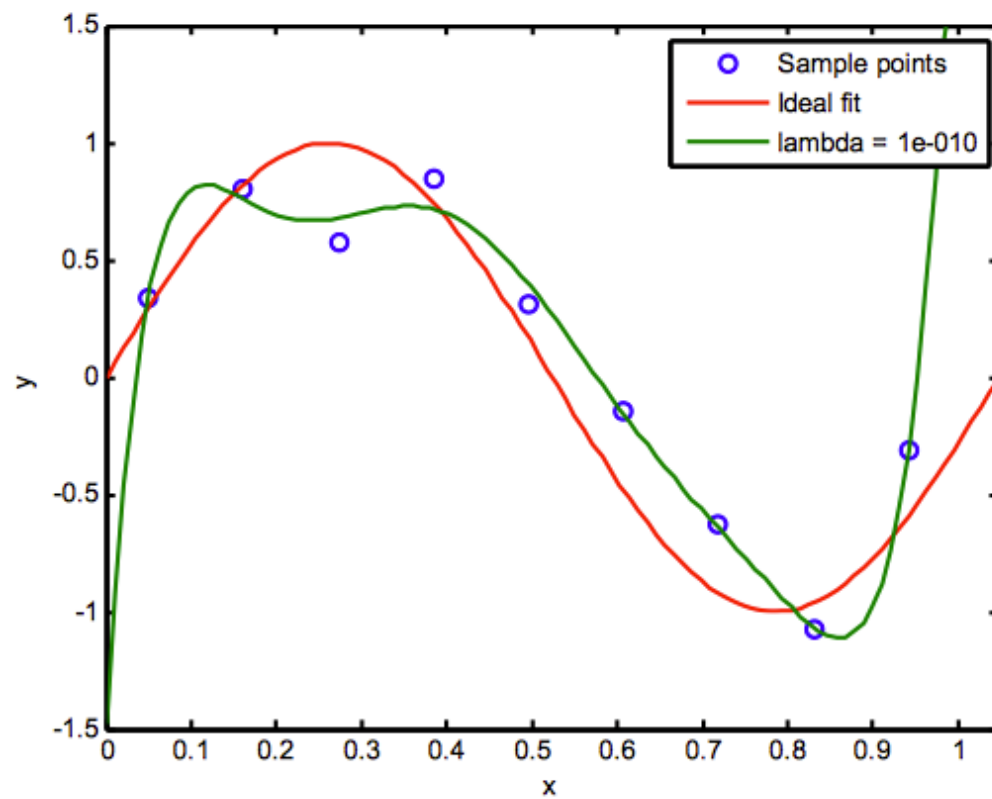
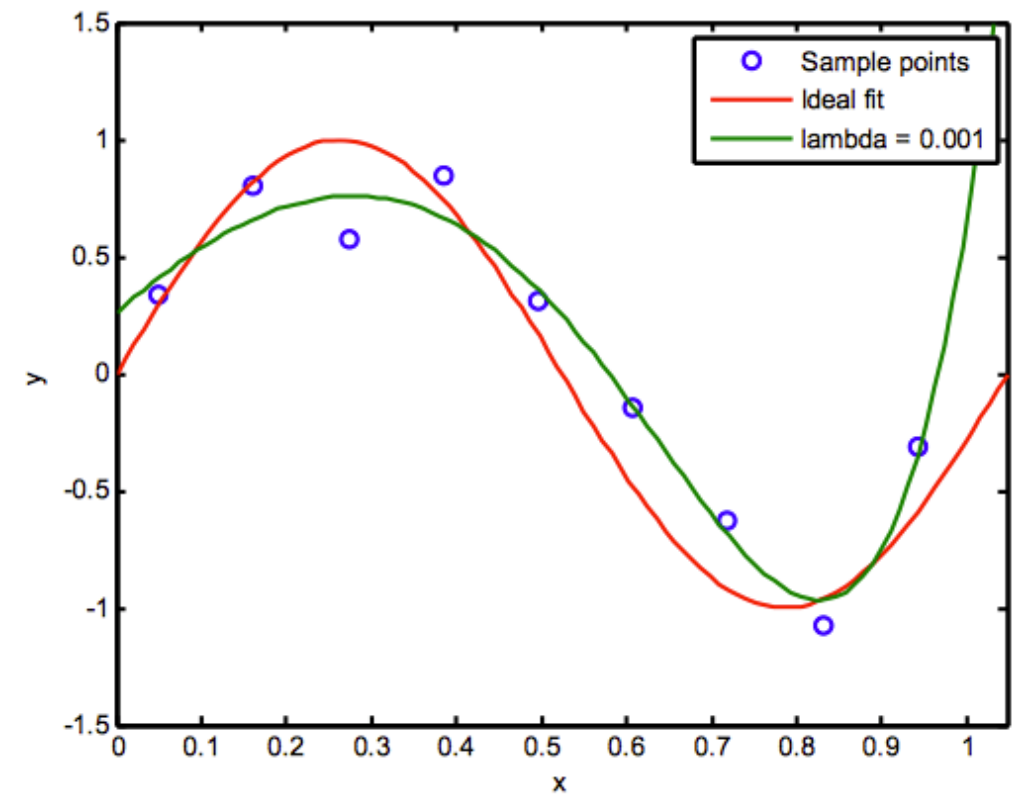
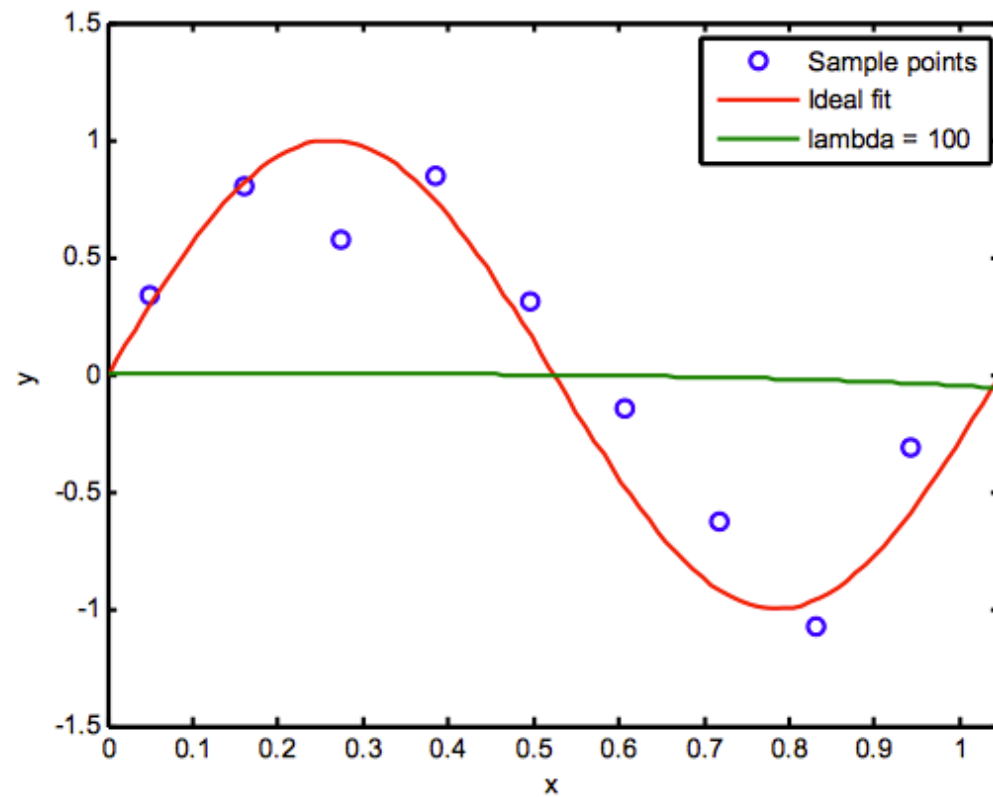


$$f(x, \theta) = z\theta$$

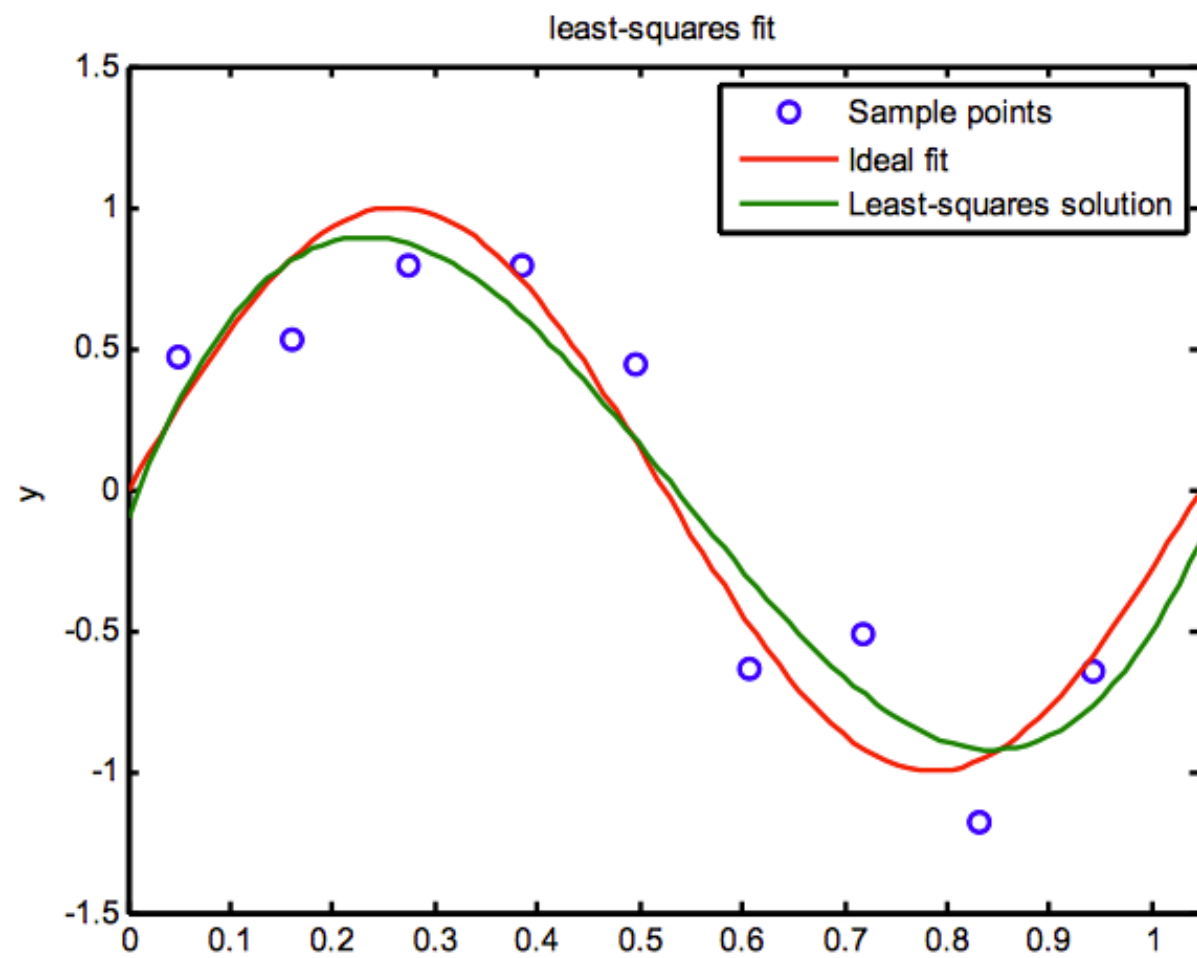
$$z: x \rightarrow z$$

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2 \quad \theta \in \mathbb{R}^{D+1}$$

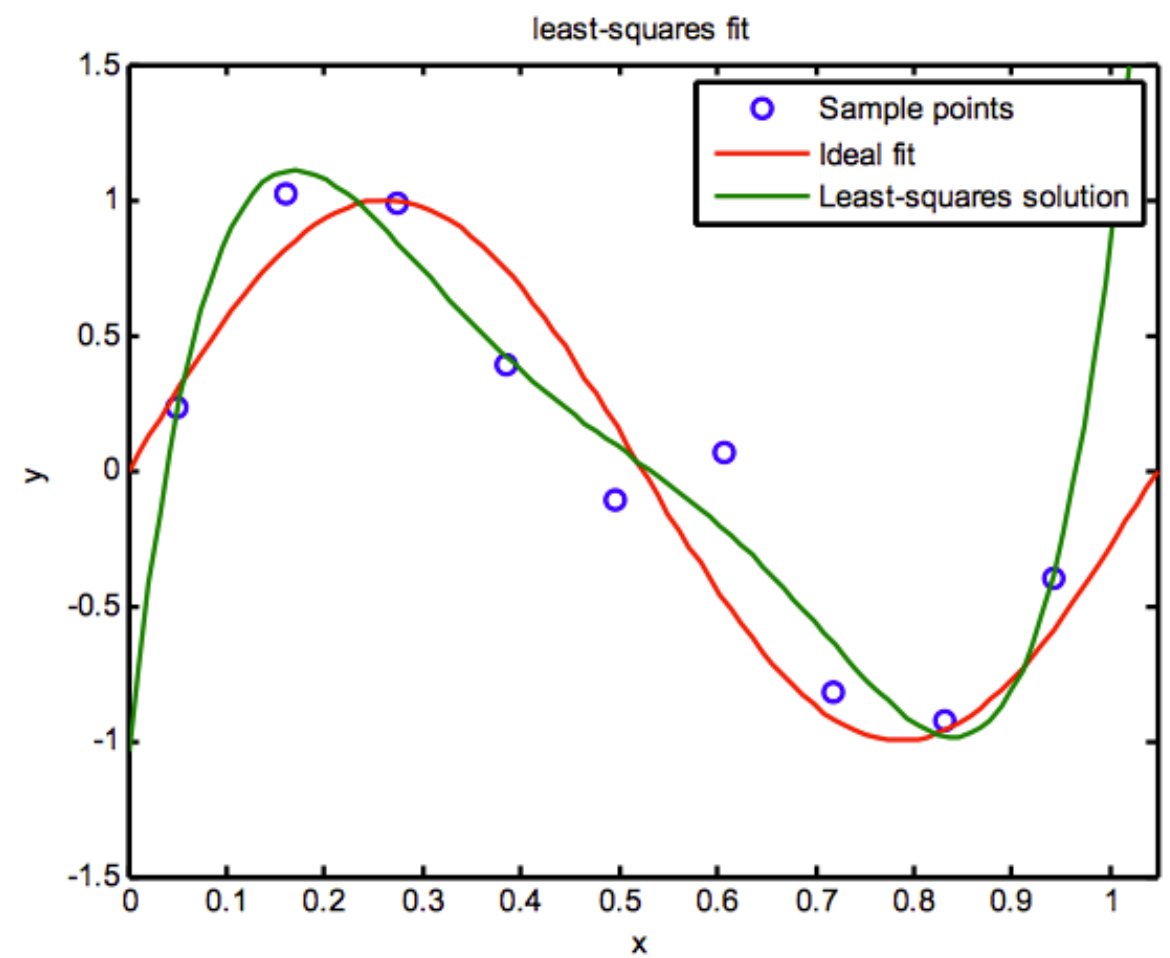
$N = 9$  samples,  $D = 7$



$D = 3$



$D = 5$



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- Ridge regression
- Lasso regression 
- Determining regularization strength

# Regularized Regression

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_2^2$$

Squared loss/Error

$$\frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2$$

L2 Regularizer

$$\lambda \|\theta\|_2^2$$

Now let's look at another regularization choice.

# The Lasso Regularization (L1 norm) and sparsity

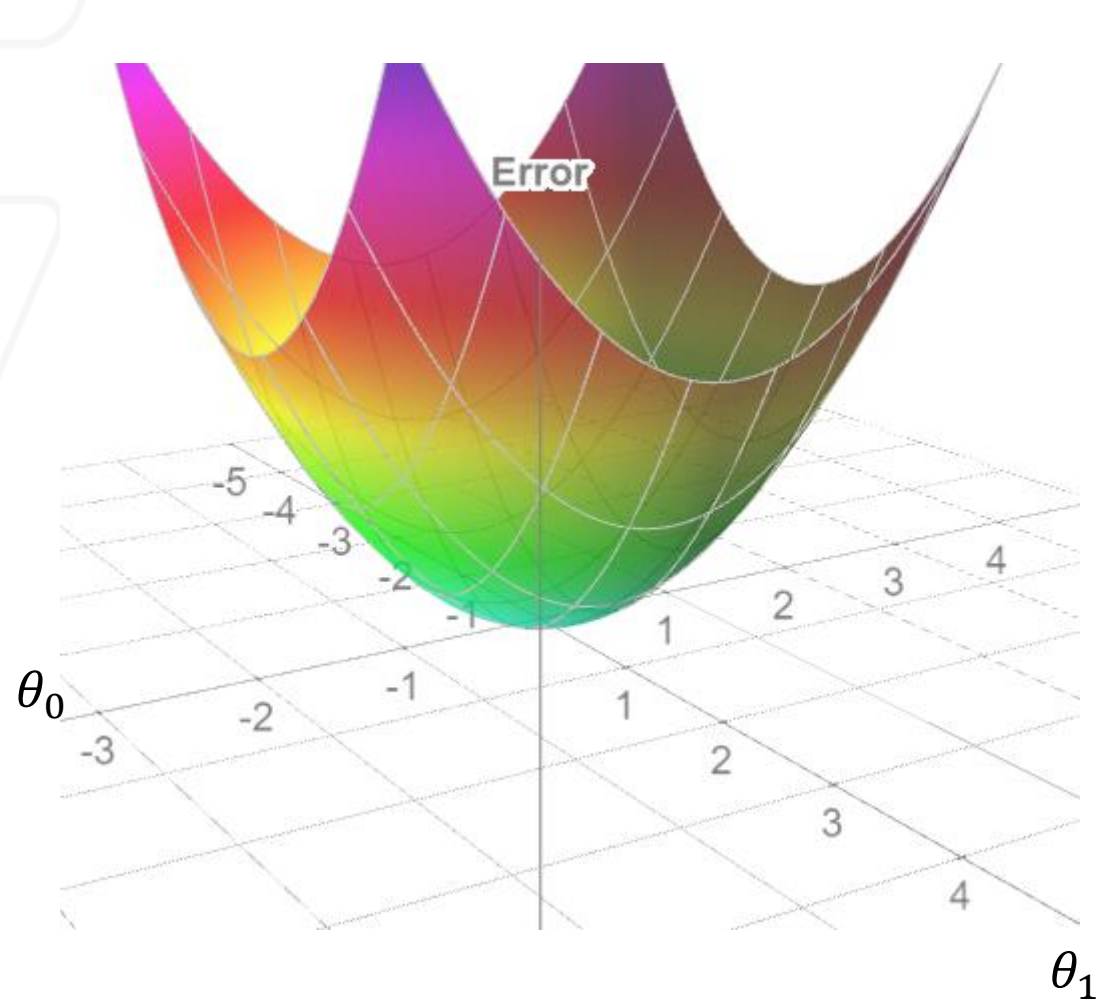
Lasso = **L**east Absolute **S**hrinkage and **S**election **O**perator

$$\tilde{E}(\theta) = \frac{1}{N} \sum_{i=1}^n (y^{\{i\}} - z^{\{i\}} \theta)^2 + \lambda \|\theta\|_1$$

L1 norm induces sparsity. This means that some of the weights become zero, and the feature contribution will be completely removed. L1 Regularizer could be used for feature selection

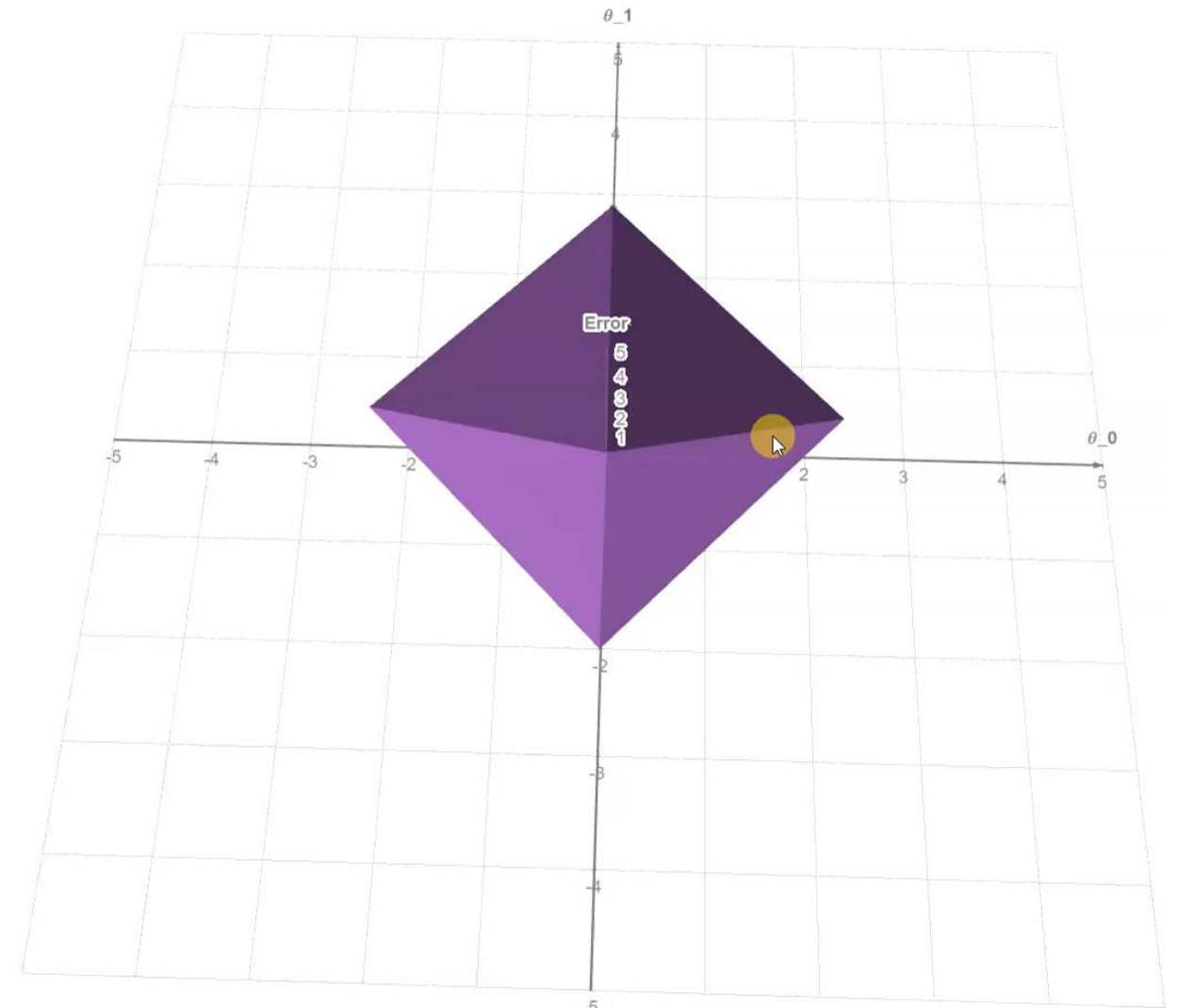
# Ridge Regularizer

$$g(\theta) = \theta_0^2 + \theta_1^2 = \theta^T \theta$$



# Lasso Regularizer

$$g(\theta) = \theta_0 + \theta_1 = \theta$$



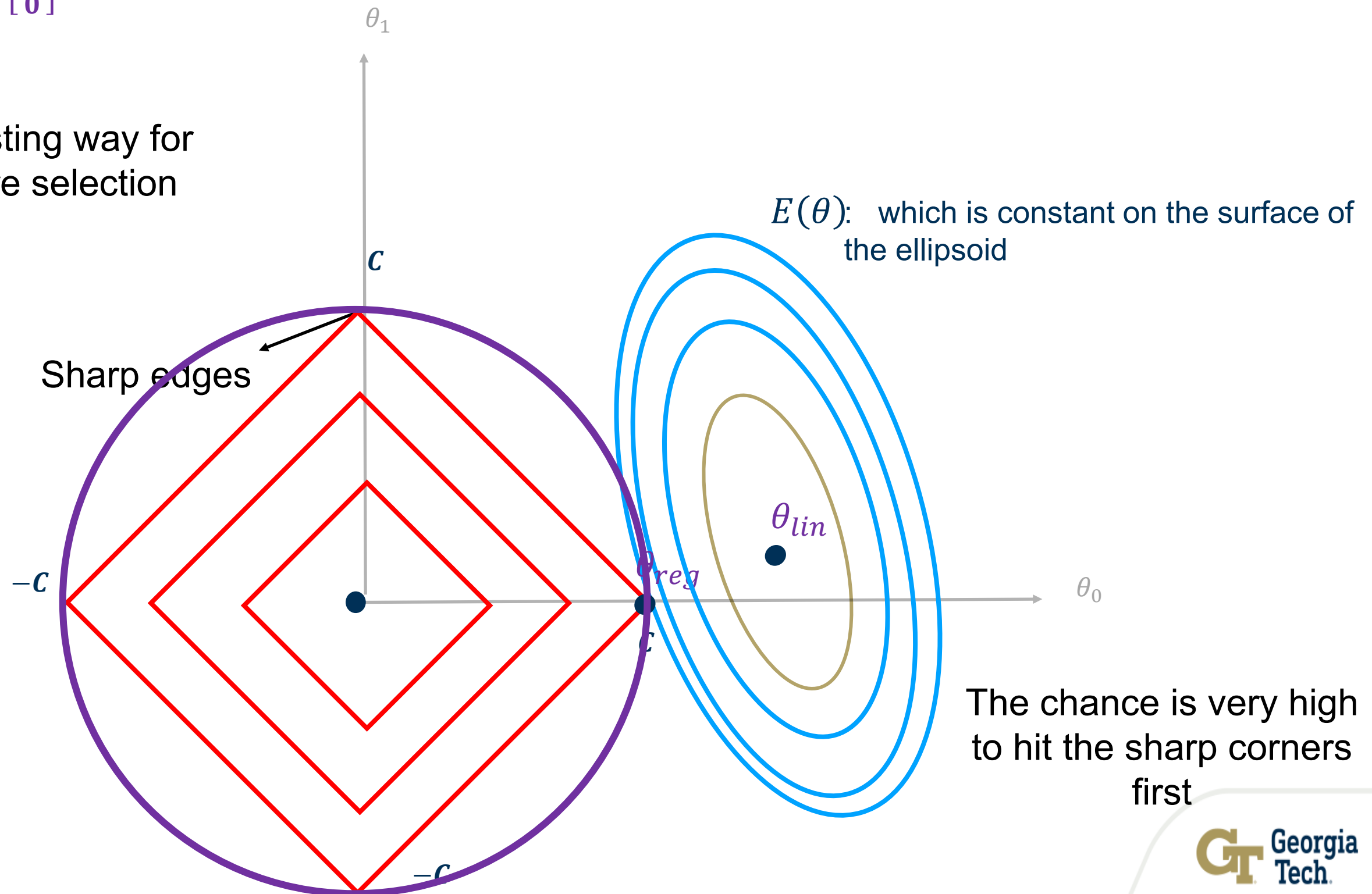
[Animation](#)

Let's say we have two parameters ( $\theta_0$  and  $\theta_1$ )

$$\text{Min } E(\theta) = \frac{1}{N} (z\theta - y)^T (z\theta - y) + \lambda \|\theta\|_1$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}$$

Interesting way for  
feature selection



# Ridge versus Lasso

## Ridge

$$\begin{aligned}\tilde{E}(\theta) \\ &= \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_2^2\end{aligned}$$

It is a convex model

Both mean squared error and L2 regularizer are differentiable.

We can get a closed form solution

## Lasso

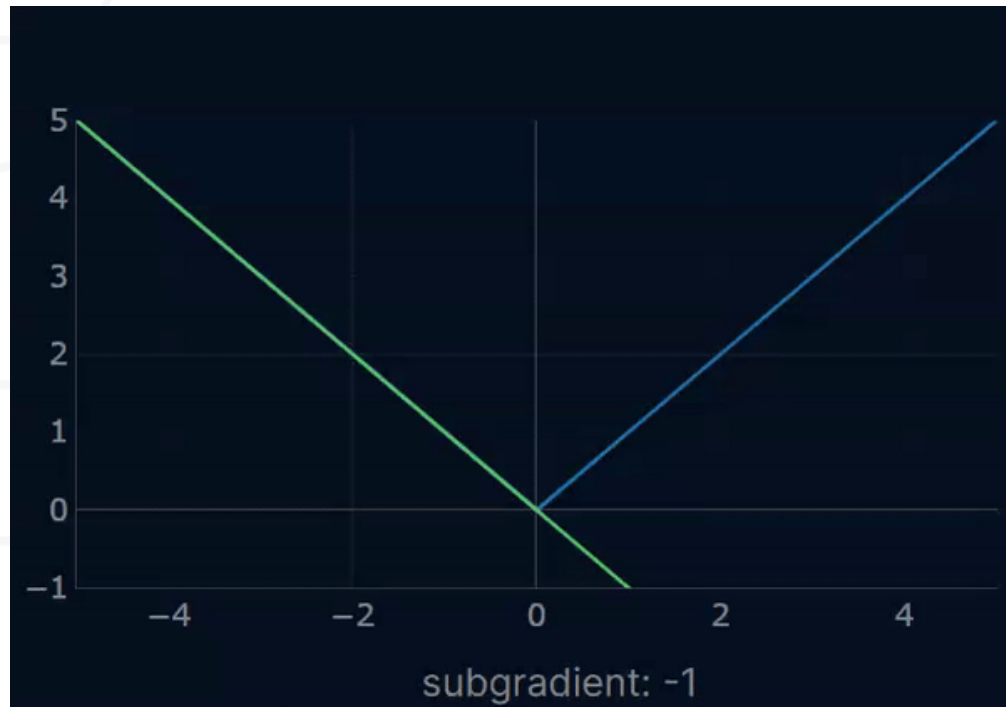
$$\begin{aligned}\tilde{E}(\theta) \\ &= \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_1\end{aligned}$$

It is a convex model

L1 regularizer is NOT differentiable.

We can **NOT** get a closed form solution

# Sub-gradient Descend in Lasso



$$\tilde{E}(\theta) = \frac{1}{N} (y - z\theta)^T (y - z\theta) + \lambda \|\theta\|_1$$

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \frac{\partial (\lambda \|\theta\|_1)}{\partial \theta}$$

Using Sub-gradient

$$\frac{\partial \tilde{E}(\theta)}{\partial \theta} = -z^T (y - z\theta) + \lambda \text{sign}(\theta)$$

In *sign* function, we use this sub-gradient line as our under-estimator (below our function)

# Recap

- Why do we need regularization?
- What is the root cause?
- How do we regularize in ridge regression?
- Why is the penalty term chosen as  $\theta^T \theta$ ?

# Recap

- Other forms of regularization covered in previous classes?
- Lasso vs Ridge Regression?
- How do we regularize in ridge regression?
- Why is the penalty term chosen as  $\theta^T \theta$ ?

| Regularization | Bias            | Variance          | Total Error | Stability          |
|----------------|-----------------|-------------------|-------------|--------------------|
| ✗<br>Without   | Low             | <b>High</b>       | High        | Overfits           |
| ✓ With         | Slightly higher | <b>Much lower</b> | Lower       | Generalizes better |

# Outline

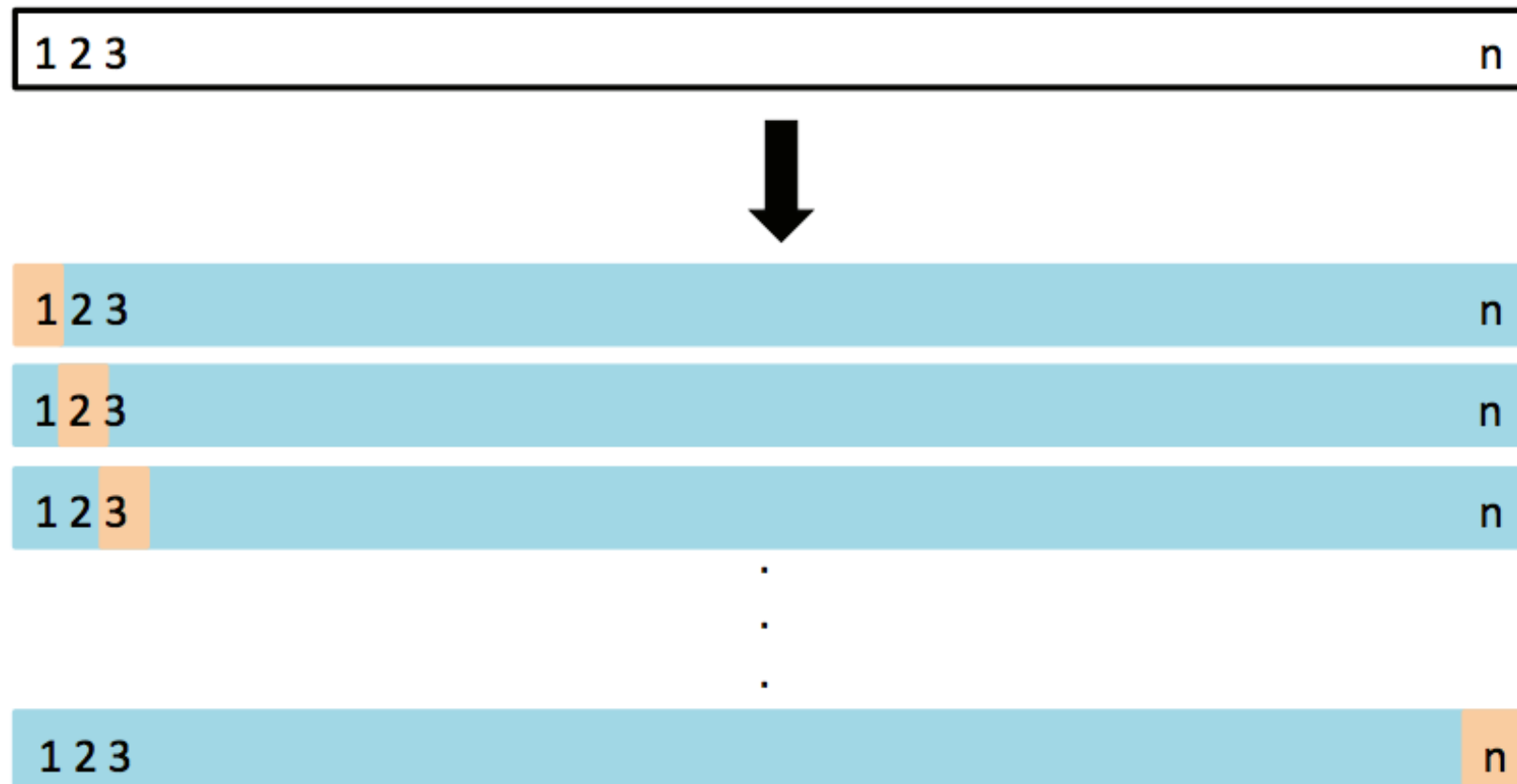
- Overfitting and regularized learning
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# Leave-One-Out Cross Validation

For every  $i = 1, \dots, n$ :

- ▶ train the model on every point except  $i$ ,
- ▶ compute the test error on the held out point.

Average the test errors.  $CV_{(n)} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i^{(-i)})^2$



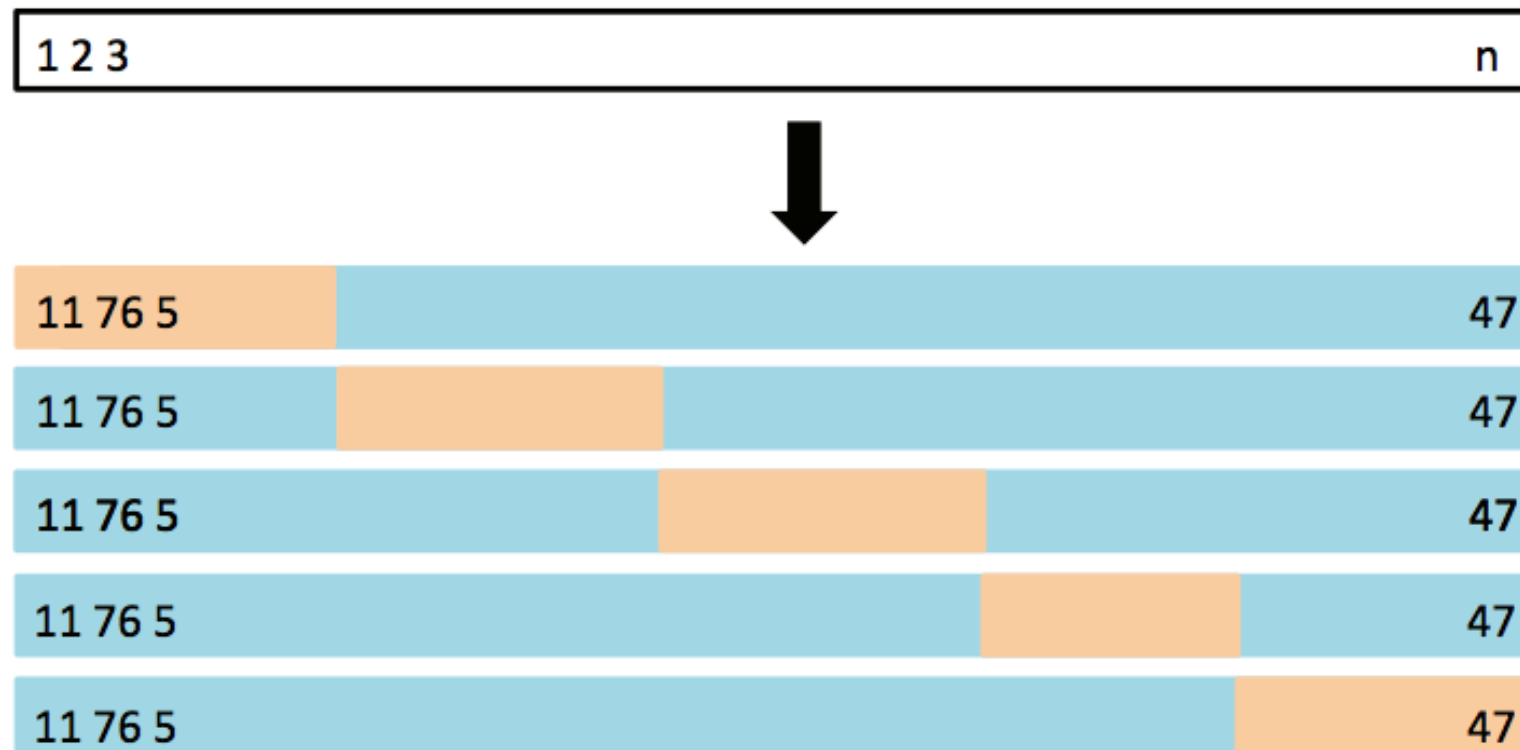
# K-Fold Cross Validation

Split the data into  $k$  subsets or *folds*.

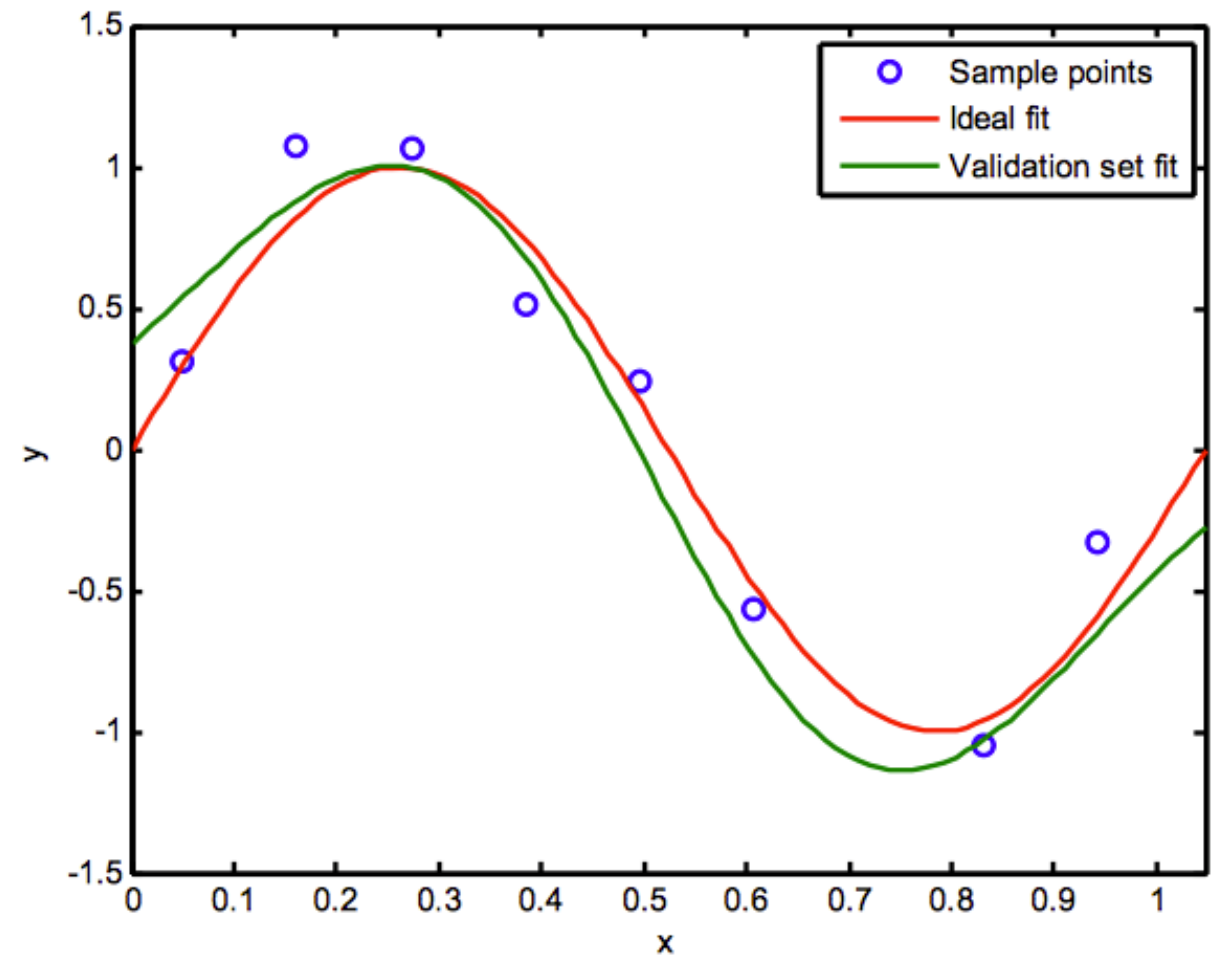
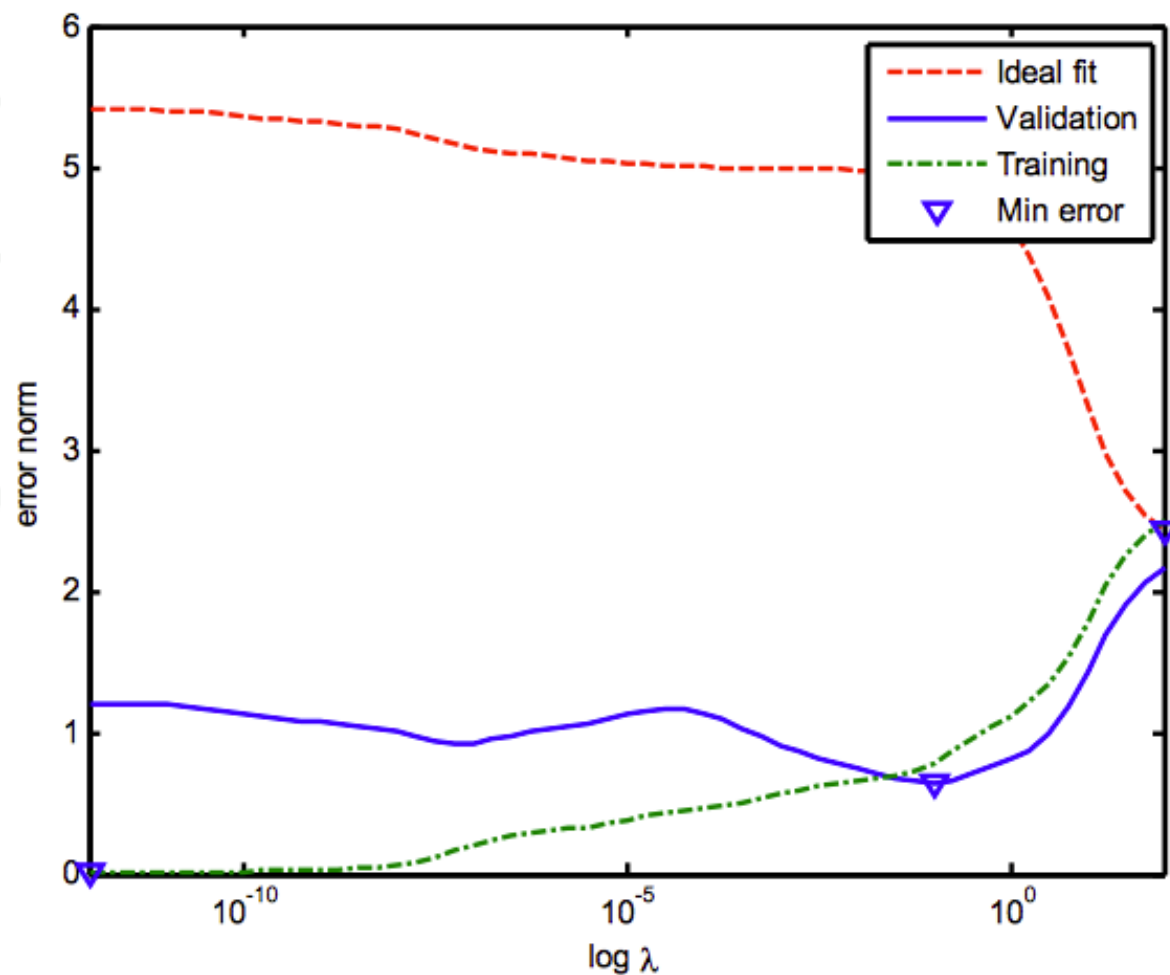
For every  $i = 1, \dots, k$ :

- ▶ train the model on every fold except the  $i$ th fold,
- ▶ compute the test error on the  $i$ th fold.

Average the test errors.



# Choosing $\lambda$ Using Validation Dataset



Pick up the lambda with the lowest mean value of rmse calculated by Cross Validation approach

# Take-Home Messages

- What is overfitting
- What is regularization
- How does Ridge regression work
- Sparsity properties of Lasso regression
- How to choose the regularization coefficient  $\lambda$

# Quick Knowledge Check